Thresholds, coherent structures, and dynamics of critical focusing dispersive systems

Concentration energy, spectral bifurcation, and transport

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Abstract

We connect critical concentration, coherent motion, and conservative evolution through five analytic results. A bounded modification of a homogeneous dispersion has an exact concentration energy determined by the high-frequency optimizer manifold. Strictly below that energy, critical-mass sublevels are uniformly bounded; a conservative local flow therefore remains bounded for all time. A concrete anisotropic quartic class in dimensions one through three includes its local flow proof. For homogeneous waves, a complementary-resolvent reduction gives the spectral pair created by small velocity, including the correction due to translation coupling. Least-mass negative-energy endpoints and regular zero-energy splitting components force degeneracy of the conserved-charge response. Internal unitary symmetry gives compactness, quantitative polarization for arbitrary fields, and profile rigidity uniformly in the number of components, including a separable Hilbert space. Finally an exact core cancellation gives a two-sided localized transport identity for all homogeneous orders $m > 1$, including anisotropic dispersion and critical Hartree interactions. Each theorem states the operator or flow assumptions it uses.

The conditional spectral assertions retain their nondegeneracy and operator-continuity hypotheses. The new global-flow theorem is unconditional in the concrete quartic class specified below.

The central connection

Critical scaling makes kinetic energy and attraction scale at the same rate. At critical mass, a concentrating sequence can keep its energy bounded only by approaching the sharp interpolation optimizer manifold. A bounded dispersion correction then survives as a finite energy on that manifold. This identifies the energy at which concentration becomes possible, and conservation turns the strict inequality below it into a statement about every time of the actual evolution.

The same coherent structures organize a second question: what becomes of the critical scaling mode when the wave starts moving? Its interaction with phase and translation modes determines the spectral growth rate. The coefficient must be computed after removing translation coupling. The third question concerns evolution that does not stay near one wave. A localized dilation identity relates sustained spatial transport to energy and momentum. These are three distinct mathematical connections; none is inferred merely from the existence of a stationary profile.

A concrete nonlinear evolution result. For $N = 1, 2, 3$, $\varepsilon > 0$, consider

$$iu_t = k(D)u - |u|^{8/N}u, \qquad k(\xi) = |\xi|^4 + \varepsilon\frac{|\xi|^{10}}{1+|\xi|^6}.$$

Let M_0 be the pure quartic sharp critical mass and put $M_\infty = (1+\varepsilon)^{N/4}M_0$. Every H^2 initial datum with

$$M(u_0) = M_\infty, \qquad E_k(u_0) < 0$$

produces a global solution with $\sup_{t\in\mathbb{R}} \|u(t)\|_{H^2} < \infty$. This class is nonempty: finite dilations of a high-frequency optimizer have strictly negative energy. The conclusion covers all such initial data, with no radial symmetry, momentum restriction, or proximity to a particular wave. It concerns this modified dispersion, whose high-frequency deficit tends to zero; it is not a blowup theorem for the pure quartic equation.

Organization and dependencies. Section 1 supplies scalar compactness and the conventions. The concentration-energy, spectral-pair, polarization, and transport arguments are given in full. The endpoint section applies to any regular profile at the defined threshold; its finite-splitting corollary uses the compactification theorem of the preceding companion manuscript [1]. The secondary scale-selection and minimizer-stability consequences explicitly identify the preceding variational inputs they import. Their source proofs are included with this manuscript's source package.

1 Analytic foundations and conventions

The Fourier transform is unitary, $D = -i\nabla$, and $P(u) = \operatorname{Im}\int \overline{u}\nabla u = \int \xi|\widehat{u}|^2$. The real Hilbert pairing is the real part of the complex L^2 pairing. Throughout the local-power arguments, $N \geq 1$, $m > 1$, $s = m/2$, $r = m/N$, and $p = 2 + 2r$. A principal symbol a is real, even, m-homogeneous, C^{N+2} off zero, and satisfies $c|\xi|^m \leq a(\xi) \leq C|\xi|^m$. Set $A(u) = \int a|\widehat{u}|^2$, $M(u) = \|u\|_2^2$, and $E(u) = A(u)/2 - \|u\|_p^p/p$. The sharp critical mass M_c is defined by

$$\|u\|_p^p \leq \frac{p}{2}\left(\frac{M(u)}{M_c}\right)^r A(u). \tag{1}$$

The traveling-wave convention $u(t,x) = e^{i\omega t}U(x - vt)$ gives $[a(D) + \omega + iv\cdot\nabla]U - |U|^{2r}U = 0$.

Lemma 1.1 (Normalized optimizer compactness)**.** *The inequality (1) is attained. If $M(f_n) \to M_c$, $A(f_n) \to 1$, and $\|f_n\|_p^p \to p/2$, then some translated subsequence converges strongly in H^s to an optimizer. The normalized optimizer set is compact modulo translations and phases.*

Proof. Sharp Sobolev interpolation gives a finite positive best constant, and amplitude and spatial scaling normalize a maximizing sequence as stated. Bounded energy-space sequences that vanish locally in mass converge to zero in L^p: use a bounded-overlap cover by unit balls, local Sobolev interpolation, and $2 < p < 2N/(N-m)$ when $N > m$ (all finite p otherwise). Thus translations give a nonzero weak limit Q and almost-everywhere convergence. Write $M_1 = M(Q) > 0$, $A_1 = A(Q) > 0$, $M_2 = M_c - M_1$, and $A_2 = 1 - A_1$. Quadratic splitting and Brezis–Lieb give

$$M_c^r \leq M_1^r A_1 + M_2^r A_2, \qquad A_1 + A_2 = 1.$$

If both masses are positive, the right side is strictly below M_c^r. Consequently $M_1 = M_c$ and then $A_1 = 1$. Weak convergence plus convergence of both quadratic norms gives strong H^s convergence. The same proof applies to a sequence of optimizers. Constant phase is proved by the density-norm lemma below; this compactness argument itself does not need a positivity-preserving kinetic operator. □

Lemma 1.2 (Weak-profile regularity and Pohozaev identities)**.** *Every weak H^s local-power profile belongs to $L^\infty \cap H^{m+1}$ and satisfies*

$$v\cdot P = 2mE, \qquad \omega M = rA + (2m - p)E. \tag{2}$$

No spatial moment is required.

Proof. Choose Λ so large that $a(\xi) - v \cdot \xi + \Lambda \asymp 1 + |\xi|^m$. The shifted resolvent maps L^q to $H^{m,q}$, $1 < q < \infty$, by the Mihlin multiplier theorem. For $N > m$, the starting exponent is $q_0 = 2N/(N-m)$. Splitting the nonlinearity into $|U| > 1$ and $|U| \le 1$, resolvent Sobolev improvement iterates the reciprocal exponent as $\beta_{j+1} = (1+2r)\beta_j - r$ with $\beta_0 = (1-r)/2$. The ideal iterates $\beta_j = 1/2 - (r/2)(1+2r)^j$ cross the boundedness threshold after finitely many steps. Inputs have exponent greater than one; intermediate exponents may be weakened to avoid endpoints. The linear contribution has the stronger improvement $\beta \mapsto \beta - r$ and is handled separately. When $N = m$ start at any sufficiently large finite exponent; when $N < m$ the initial space already embeds in L^∞. Now the nonlinearity is in L^2, giving $U \in H^m \subset H^1$. The C^1 map $z \mapsto |z|^{2r} z$, with bounded derivative on the range of U, sends U into H^1; one more resolvent application gives H^{m+1}.

Here are the cutoff details for the dilation calculation. Let $f_R(x) = x\chi(x/R)$ with χ compactly supported and equal to one near zero, and let $L_R = f_R \cdot \nabla + \operatorname{div} f_R/2$. The Fourier kernel of $[a(D), L_R]$ is, up to the transform constant,

$$\frac{i}{2}(a(\xi) - a(\eta))(\xi + \eta) \cdot \widehat{f}_R(\xi - \eta).$$

Splitting comparable and noncomparable frequencies gives

$$|a(\xi) - a(\eta)||\xi + \eta| \le C|\xi - \eta|(|\xi| + |\eta|)^m.$$

After division by $\langle\xi\rangle^s\langle\eta\rangle^s$, the convolution kernel is bounded by $C|k|\langle k\rangle^s R^{N+1}|\widehat{f}(Rk)|$, with a uniform L^1 norm for $R \ge 1$. Hence the commutators are uniformly bounded $H^s \to H^{-s}$. Schwartz approximation and the homogeneous symbol identity give $\langle U, [a(D), L_R]U\rangle \to mA(U)$. The drift commutator tends to $-v \cdot P$; its order is one and $s > 1/2$. In the nonlinear pairing integration by parts and dominated convergence give

$$\operatorname{Re} \int |U|^{p-2}\overline{U} L_R U \longrightarrow N(1/2 - 1/p)\|U\|_p^p = \frac{m}{p}\|U\|_p^p.$$

Testing the profile with $L_R U$ therefore gives $mE = v \cdot P/2$. Testing with U gives $A + \omega M - v \cdot P - \|U\|_p^p = 0$; combining these two equations proves (2). □

The same regularity proof holds for an individually fixed nonnegative order-m addition b with the stated Mihlin derivative bounds. Constants need not be uniform in that addition or in v. For nonhomogeneous minimizers, differentiation of their scalar mass-preserving dilation energy is legitimate directly by dominated convergence and minimality; differentiability of the dilation orbit as an H^s-valued map is not being assumed.

2 Concentration energy and global dynamics at critical mass

A high-frequency concentration has a definite limiting energy contribution from a bounded correction to the homogeneous dispersion. We identify that contribution using the full optimizer manifold. Conservation then converts the resulting compactness statement into a global theorem for all initial data below the concentration energy, including data at the critical mass. No proximity to a standing wave is needed.

2.1 The homogeneous problem and the concentration energy

Fix $N \ge 1$, $m > 1$, $s = m/2$, $r = m/N$, and $p = 2 + 2r$. Let a be real, even, positive elliptic, and homogeneous of degree m. Write

$$M(f) = \|f\|_2^2, \qquad A(f) = \int a(\xi)|\widehat{f}(\xi)|^2\, d\xi, \qquad E_a(f) = \tfrac{1}{2}A(f) - \tfrac{1}{p}\|f\|_p^p.$$

Let M_c be the sharp mass in

$$\frac{1}{p}\|f\|_p^p \le \frac{1}{2}\left(\frac{M(f)}{M_c}\right)^r A(f), \tag{3}$$

and define the normalized equality manifold

$$\mathcal{G} = \{Q \in H^s : M(Q) = M_c,\ A(Q) = 1,\ E_a(Q) = 0\}.$$

We use the normalized sharp-GN compactness property: whenever $M(f_n) \to M_c$, $A(f_n) \to 1$, and $E_a(f_n) \to 0$, there are translations and a subsequence converging strongly in H^s to an element of $\mathcal{G}$. For the homogeneous symbols considered in this work this is the established sharp-GN compactness input. It requires neither optimizer uniqueness nor nondegeneracy of the real linearized operator. In particular $\mathcal{G}$ is compact modulo translations; constant phases are already included in this set.

Let d be a bounded, measurable, real, even Fourier multiplier. Assume that there is a bounded function d_∞ on S^{N-1} such that

$$\sup_{\theta\in S^{N-1}} |d(R\theta) - d_\infty(\theta)| \longrightarrow 0 \qquad (R \longrightarrow \infty). \tag{4}$$

Define

$$k = a - d, \qquad D(f) = \int d(\xi)|\widehat{f}(\xi)|^2\,d\xi, \qquad E_k(f) = E_a(f) - \tfrac{1}{2}D(f).$$

No sign assumption is imposed on d or d_∞. The quadratic perturbation is continuous on L^2, and E_k is continuous on H^s. Set

$$e_\infty = -\frac{1}{2}\max_{Q\in\mathcal{G}} \int d_\infty(\xi/|\xi|)|\widehat{Q}(\xi)|^2\,d\xi. \tag{5}$$

The maximum exists: its objective is a bounded-multiplier quadratic form, continuous in L^2 and invariant under translations and constant phase. Compactness of $\mathcal{G}$ modulo translations applies. Since the limiting multiplier has degree zero, the value does not depend on the kinetic normalization used for the equality manifold.

Theorem 2.1 (Energy of a concentrating sequence)**.** *Suppose*

$$M(f_n) \le M_c, \qquad A(f_n) \longrightarrow \infty, \qquad \sup_n E_k(f_n) < \infty.$$

Then $M(f_n) \to M_c$. Put $t_n = A(f_n)^{1/m}$ and write $f_n(x) = t_n^{N/2}V_n(t_n x)$. After extraction and translations, $V_n \to Q$ strongly in H^s for some $Q \in \mathcal{G}$, and

$$D(f_n) \longrightarrow \int d_\infty(\xi/|\xi|)|\widehat{Q}(\xi)|^2\,d\xi, \tag{6}$$

$$\liminf_n E_k(f_n) \ge -\frac{1}{2}\int d_\infty(\xi/|\xi|)|\widehat{Q}(\xi)|^2\,d\xi \ge e_\infty. \tag{7}$$

Proof. The sharp inequality and the mass bound give $E_a(f_n) \ge 0$. Boundedness of d gives

$$0 \le E_a(f_n) = E_k(f_n) + \tfrac{1}{2}D(f_n) \le \sup_j E_k(f_j) + \tfrac{1}{2}\|d\|_\infty M_c.$$

Consequently

$$A(V_n) = 1, \qquad E_a(V_n) = t_n^{-m}E_a(f_n) \longrightarrow 0.$$

Equation (3) also gives

$$0 \le \frac{1}{2}\left[1 - \left(\frac{M(f_n)}{M_c}\right)^r\right] \le E_a(V_n) \longrightarrow 0.$$

Thus $M(f_n) \to M_c$, and normalized sharp-GN compactness yields the asserted strong convergence. Equivalently one may first correct the masses and kinetic normalizations by factors tending to one and then undo those normalizations after obtaining a compact subsequence.

Translations do not change Fourier modulus. Since d is bounded, strong L^2 convergence implies

$$\left| \int d(t_n\xi)(|\widehat{V}_n(\xi)|^2 - |\widehat{Q}(\xi)|^2)\, d\xi \right| \le \|d\|_\infty \|V_n - Q\|_2 (\|V_n\|_2 + \|Q\|_2) \longrightarrow 0.$$

For every $\xi \neq 0$, (4) gives $d(t_n\xi) \to d_\infty(\xi/|\xi|)$. Dominated convergence now proves (6). Finally $E_k(f_n) = E_a(f_n) - D(f_n)/2 \ge -D(f_n)/2$, proving (7). □

Corollary 2.2 (Bounded energy sublevels and sharpness)**.** *For every $e < e_\infty$,*

$$\sup\{A(f) : M(f) \le M_c,\ E_k(f) \le e\} < \infty, \tag{8}$$

where the assertion is vacuous if the set is empty. Conversely, for every $e > e_\infty$ the exact-mass sublevel $\{f : M(f) = M_c,\ E_k(f) \le e\}$ is unbounded in H^s.

Proof. Failure of (8) contradicts Theorem 2.1. For sharpness choose $Q \in \mathcal{G}$ attaining the maximum in (5) and let $Q_\lambda(x) = \lambda^{N/2} Q(\lambda x)$. Homogeneity gives

$$M(Q_\lambda) = M_c, \qquad A(Q_\lambda) = \lambda^m, \qquad E_a(Q_\lambda) = 0.$$

Moreover

$$E_k(Q_\lambda) = -\tfrac{1}{2} \int d(\lambda\xi) |\widehat{Q}(\xi)|^2\, d\xi \longrightarrow e_\infty \qquad (\lambda \longrightarrow \infty).$$

For any fixed $e > e_\infty$ these trials eventually belong to the stated sublevel while their kinetic energies diverge. □

The strict comparison in this result is optimal for mass–energy coercivity. At $e = e_\infty$, boundedness depends on the particular symbol. Sharpness of an energy sublevel does not itself construct a blow-up trajectory.

Remark 2.3 (Choice of the zero of frequency). Replacing k by $k + c$ multiplies solutions by e^{-ict}, adds $cM/2$ to their energy, and adds $cM_c/2$ to e_∞. Thus the difference $E_k - e_\infty$ at $M = M_c$ is unchanged. The concentration-energy comparison does not depend on the arbitrary zero of frequency.

2.2 Conservation turns the bound into global dynamics

For the general-symbol statement, we specify the local evolution input. Let (F) denote the following properties of

$$i\partial_t u = k(D)u - |u|^{p-2}u : \tag{9}$$

for every $u_0 \in H^s$ there is a unique maximal solution $u \in C((T_-, T_+), H^s)$, depending continuously on its initial datum, conserving M and E_k, and satisfying the continuation alternative that a finite lifespan endpoint forces divergence of the H^s norm. Ellipticity of an arbitrary symbol is not by itself being used as a substitute for this local evolution theory.

Theorem 2.4 (Global bounded evolution at critical mass)**.** *Assume* (F). *If*

$$M(u_0) \le M_c, \qquad E_k(u_0) < e_\infty,$$

then the solution of (9) *is global in both time directions and*

$$\sup_{t \in \mathbb{R}} \|u(t)\|_{H^s} < \infty.$$

For every strictly subcritical datum $M(u_0) < M_c$, the same conclusion holds without any restriction on its energy.

Proof. In the first case, conservation places the maximal orbit in the fixed sublevel of Corollary 2.2 with $e = E_k(u_0) < e_\infty$. Ellipticity of a and conservation of mass bound its H^s norm. The continuation alternative excludes both finite endpoints. For the second assertion the direct estimate

$$E_k(f) \geq \frac{1}{2}\left[1 - \left(\frac{M(f)}{M_c}\right)^r\right] A(f) - \frac{1}{2}\|d\|_\infty M(f) \tag{10}$$

is coercive at each fixed mass below M_c; conservation gives the same conclusion. □

Proposition 2.5 (A concrete local theory including anisotropy)**.** *For $m = 4$ and $N \in \{1, 2, 3\}$, hypothesis* (F) *holds for every symbol $k = a - d$ in this section. Consequently Theorem 2.4 is unconditional for these quartic equations, including anisotropic a.*

Proof. Here $s = 2$ and the nonlinear map is $F(z) = |z|^\alpha z$ with $\alpha = 8/N > 2$. Its second derivative, viewing $\mathbb{C}$ as $\mathbb{R}^2$, is locally Lipschitz. The embeddings $H^2(\mathbb{R}^N) \hookrightarrow L^\infty \cap W^{1,4}$ and the ordinary chain and product rules imply, on every bounded H^2 ball,

$$\|F(u) - F(v)\|_{H^2} \leq C_R \|u - v\|_{H^2} \quad \text{if } \|u\|_{H^2} + \|v\|_{H^2} \leq R. \tag{11}$$

For example the second-derivative terms have the forms $F'(u)D^2u$ and $F''(u)(Du)^2$; their differences are bounded using $H^2 \hookrightarrow L^\infty$, $H^2 \hookrightarrow W^{1,4}$, and the local Lipschitz bounds on F' and F''.

Since k is real, $e^{-itk(D)}$ is a strongly continuous unitary group on H^2. Its Duhamel map and (11) give a contraction in $C([-T, T], H^2)$ with T depending only on the initial H^2 norm. This proves existence, uniqueness, continuous dependence, and the H^2 continuation alternative.

To justify conservation without assuming excessive smoothness of the non-polynomial power, let P_R be the orthogonal Fourier projection to $\{\xi : |\xi| \leq R\}$ and solve

$$i\partial_t u_R = k(D)u_R - P_R F(u_R), \qquad u_R(0) = P_R u_0.$$

The band-limited subspace is preserved. On that subspace $k(D)$ is a bounded operator on H^2, so the solution is differentiable in H^2. The real L^2 pairings defining the derivatives of mass and energy are therefore legitimate. If $g_R = k(D)u_R - F(u_R)$, then $\partial_t u_R = -iP_R g_R$, so

$$\frac{d}{dt} E_k(u_R) = \mathrm{Re}\langle g_R, -iP_R g_R\rangle = 0.$$

Likewise $dM(u_R)/dt = 0$, since $P_R u_R = u_R$ and $\int \overline{u_R} F(u_R)$ is real. The local contraction bounds are uniform in R. Strong convergence $P_R \to \mathrm{Id}$ on H^2 and the Duhamel equations yield $u_R \to u$ in $C([-T, T], H^2)$. Continuity of M and E_k passes conservation to u. Iterating over the maximal lifespan proves (F). □

2.3 Consequences for the two-threshold interpolation

Let a_0 and a_∞ be positive elliptic real-even symbols of degree m, with $h = a_\infty - a_0$ positive elliptic. Consider

$$\begin{aligned} k_q(\xi) &= a_0(\xi) + h(\xi)\frac{|\xi|^q}{1 + |\xi|^q} = a_\infty(\xi) - d_q(\xi), \\ d_q(\xi) &= \frac{h(\xi)}{1 + |\xi|^q}, \qquad q > 0. \end{aligned} \tag{12}$$

Denote the two homogeneous sharp masses by $M_0 < M_\infty$. The following existence and compactness inputs are imported from the companion study *Thresholds, spectral geometry, and persistent radiation in mass-critical dispersive equations*, Section 5; they are not reproved in the present dynamical argument. That analysis gives $I(M) = \inf_{M(f)=M} E_{k_q}(f) < 0$ with attainment for $M_0 < M < M_\infty$. At $M = M_\infty$, the infimum is also negative and attained when $q > m$, and every minimizing sequence is compact in H^s modulo translations.

Corollary 2.6 (All negative-energy endpoint data are bounded)**.** *Assume a conservative local flow with the properties* (F) *for k_q. For every $q > 0$, all initial data of mass less than M_∞ generate global uniformly H^s-bounded solutions. If $q > m$, the same conclusion holds for every initial datum satisfying*

$$M(u_0) = M_\infty, \qquad E_{k_q}(u_0) < 0. \tag{13}$$

For $m = 4$, $N = 1, 2, 3$, and $q > 4$, the latter assertion requires no additional local-flow hypothesis.

Proof. For all $q > 0$, and every $\eta > 0$, homogeneity gives a finite C_η such that $k_q \geq (1-\eta)a_\infty - C_\eta$. Hence

$$E_{k_q}(f) \geq \frac{1}{2}\left[1 - \eta - \left(\frac{M(f)}{M_\infty}\right)^r\right] A_\infty(f) - \frac{1}{2} C_\eta M(f).$$

For a fixed subcritical mass choose η to make the coefficient positive and use conservation. If $q > m$, d_q is bounded and tends uniformly to zero at high frequency. Thus $d_\infty = 0$ and $e_\infty = 0$, so Theorem 2.4 proves (13). The final assertion follows from Proposition 2.5. □

The endpoint result covers arbitrary data in the stated class: it uses no radial symmetry, momentum restriction, or small distance to a standing wave. It excludes both finite-time blow-up and unbounded infinite-time growth of the energy-space norm.

For $q = m$, the bounded correction has angular limit $h(\theta)$, so (5) supplies the precise endpoint comparison. If $I(M_\infty) < e_\infty$, its strict sublevel is nonempty and Theorem 2.4 applies there. In the isotropic case $h(\xi) = \varepsilon|\xi|^m$,

$$e_\infty = -\tfrac{1}{2}\varepsilon M_\infty, \qquad E_{k_m}(f) > -\tfrac{1}{2}\varepsilon M_\infty \quad \text{for every nonzero } f \text{ of mass } M_\infty.$$

Thus that strict sublevel is empty; no endpoint global conclusion is being inferred in the isotropic borderline case. For $q < m$ the correction is unbounded at high frequency and the concentration theorem above does not apply. The variational equality $I(M_\infty) = -\infty$ in that case does not itself prove finite-time blow-up.

For $q > m$, if Q is an a_∞ equality profile, then $E_{k_q}(Q_\lambda) < 0$ for every finite λ, while $E_{k_q}(Q_\lambda) \to 0$ and $A_\infty(Q_\lambda) \to \infty$. A fixed negative energy therefore gives a bound, but no uniform bound can cover all negative energies approaching zero.

2.4 Stability of minimizing sets and the critical mass boundary

At a mass c where the constrained infimum is attained define

$$\Sigma_c = \{U \in H^s : M(U) = c,\ E_k(U) = I(c)\}.$$

This entire set includes translations and constant phases. Its stability makes no uniqueness assertion about individual profiles.

Theorem 2.7 (Stability in the correct perturbation class)**.** *Assume* (F).

1. *For the interpolation* (12)*, if $M_0 < c < M_\infty$, then Σ_c is orbitally stable under all sufficiently small H^s perturbations.*
2. *For a bounded correction as above, suppose $c = M_c$, $I(c) < e_\infty$, and every minimizing sequence at mass c is precompact in H^s modulo translations. Then Σ_c is orbitally stable under sufficiently small H^s perturbations whose masses are at most c. In particular it is stable on the exact mass surface $M = c$. This applies to* (12) *at $c = M_\infty$ whenever $q > m$.*

In both assertions all sufficiently close initial data in the indicated perturbation class generate global solutions. More explicitly, for any $\epsilon > 0$ there is $\delta > 0$ such that $\operatorname{dist}_{H^s}(u_0, \Sigma_c) < \delta$ implies

$$\sup_{t \in \mathbb{R}} \operatorname{dist}_{H^s}(u(t), \Sigma_c) < \epsilon,$$

with the additional mass restriction in the second assertion.

Proof. The compactness assumption implies that Σ_c is bounded in H^s. Thus closeness to this set makes the mass and energy uniformly close to c and $I(c)$, respectively. In the first assertion all sufficiently close data have mass uniformly below M_∞; the subcritical coercivity estimate used in Corollary 2.6 makes their solutions global and bounds their H^s norms uniformly. In the second assertion choose e strictly between $I(c)$ and e_∞ and apply Corollary 2.2 to masses at most c and energies at most e. This gives the same global existence and uniform bound.

If stability failed, there would be initial data $u_{0,n}$ with $\operatorname{dist}_{H^s}(u_{0,n}, \Sigma_c) \to 0$ and times t_n such that $f_n = u_n(t_n)$ stays a fixed positive distance from Σ_c. Conservation gives

$$M(f_n) \longrightarrow c, \qquad E_k(f_n) \longrightarrow I(c),$$

and the preceding estimates bound f_n in H^s. Set $g_n = (c/M(f_n))^{1/2} f_n$. Then

$$M(g_n) = c, \qquad \|g_n - f_n\|_{H^s} \longrightarrow 0, \qquad E_k(g_n) \longrightarrow I(c).$$

Compactness of exact-mass minimizing sequences now gives

$$\operatorname{dist}_{H^s}(g_n, \Sigma_c) \longrightarrow 0.$$

The same follows for f_n, a contradiction. □

The proof is the established variational compactness mechanism for orbital stability. The additional concentration-energy bound is what allows it to operate at the critical mass. At that boundary the restriction $M(u_0) \le M_c$ cannot be silently omitted: nearby data can otherwise cross into the supercritical mass range, where the argument supplies no uniform kinetic bound. Neither unconstrained endpoint stability nor instability is asserted here.

2.5 Removal of the flat scale direction

The interpolation also produces an exact measure of the energy cost of changing the spatial scale of a minimizer. This statement does not require spectral nondegeneracy.

Proposition 2.8 (Exact scale curvature)**.** *Let $U \neq 0$ be a constrained energy minimizer for (12). Define*

$$U_t(x) = e^{Nt/2} U(e^t x), \qquad \mathcal{K}(U) = \frac{d^2}{dt^2} E_{k_q}(U_t)\Big|_{t=0}.$$

Then the scalar energy curve is twice differentiable and, with $z = |\xi|^q$,

$$\mathcal{K}(U) = \frac{q}{2} \int h(\xi) \frac{z[(m+q) + (m-q)z]}{(1+z)^3} |\widehat{U}(\xi)|^2 \, d\xi. \tag{14}$$

For $0 < q \le m$, $\mathcal{K}(U) > 0$. For $q > m$, minimality still gives $\mathcal{K}(U) \ge 0$, but positivity does not follow pointwise from the integrand.

Proof. Let $b(\xi) = h(\xi)|\xi|^q/(1 + |\xi|^q)$ and let $\mathcal{D} = \xi \cdot \nabla_\xi$. The derivatives $\mathcal{D}^j b$, $j = 0, 1, 2$, are bounded in absolute value by $C|\xi|^m$. Therefore differentiating the scalar Fourier energy is justified by dominated convergence, with no assumption of a spatial moment or differentiability of $t \mapsto U_t$ in H^s. Homogeneity gives

$$E_{k_q}(U_t) = e^{mt} E_{a_0}(U) + \frac{1}{2} \int b(e^t \xi) |\widehat{U}(\xi)|^2 \, d\xi.$$

Its first derivative vanishes at the scalar minimum $t = 0$. Eliminating $E_{a_0}(U)$ from the second derivative yields

$$\mathcal{K}(U) = \frac{1}{2} \int (\mathcal{D}^2 - m\mathcal{D}) b(\xi) |\widehat{U}(\xi)|^2 \, d\xi.$$

Since $\mathcal{D}h = mh$ and $\mathcal{D}z = qz$, direct differentiation gives

$$(\mathcal{D}^2 - m\mathcal{D})b = qh\frac{z[(m+q)+(m-q)z]}{(1+z)^3},$$

proving (14). If $q \le m$, its coefficient is strictly positive for every $\xi \neq 0$, and a nonzero L^2 function cannot have all Fourier mass at the origin. If $q > m$, the coefficient changes sign at $z = (m+q)/(q-m)$; nonnegativity of the integral follows instead from minimality. □

For the homogeneous critical optimizer the energy is exactly flat under mass-preserving dilation. Proposition 2.8 proves that the rational correction removes this degeneracy in the scale direction when $q \le m$. It does not identify a complete Hamiltonian spectral mode.

We give the precise lower-threshold consequence of the selection theorem established for this interpolation. Put

$$\alpha = \frac{M}{M_0}, \qquad d = \alpha^r - 1, \qquad U_M(x) = \sqrt{\alpha}\,\lambda_M^{N/2} V_M(\lambda_M x),$$
$$M(V_M) = M_0, \qquad A_0(V_M) = 1.$$

Let

$$\mathcal{G}_0 = \{Q : M(Q) = M_0,\ A_0(Q) = 1,\ E_{a_0}(Q) = 0\},$$
$$C(Q) = \int h(\xi)|\xi|^q |\widehat{Q}(\xi)|^2\, d\xi,$$

and suppose at least one element of $\mathcal{G}_0$ has finite C. Write $C_{\min} = \min_{\mathcal{G}_0} C$ and

$$\beta_*^q = \frac{m}{(m+q)C_{\min}}, \qquad K_* = \frac{q}{2(m+q)}\beta_*^m.$$

The lower-threshold selection theorem of the same companion study provides the following inputs; the new deduction here is the curvature asymptotic, not a new proof of profile selection:

$$\lambda_M \sim \beta_* d^{1/q}, \qquad I(M) \sim -\alpha K_* d^{1+m/q}, \tag{15}$$

$$\frac{B(U_M)}{A_0(U_M)} = \frac{m}{m+q}d + o(d), \qquad \int \frac{h(\xi)|\xi|^q}{1+|\lambda_M\xi|^q}|\widehat{V}_M(\xi)|^2\, d\xi \longrightarrow C_{\min}, \tag{16}$$

where $B(f) = \int b|\widehat{f}|^2$. After translations and extraction, V_M converges strongly in H^s to some $Q \in \mathcal{G}_0$ with $C(Q) = C_{\min}$. The correction-energy convergence in (16), not merely strong H^s convergence, is needed in the next deduction.

Corollary 2.9 (Sharp lower-threshold scale curvature)**.** *For minimizers satisfying the preceding selection conclusions, for every $q > 0$,*

$$\mathcal{K}(U_M) \sim \frac{mq}{2} d A_0(U_M), \qquad \frac{\mathcal{K}(U_M)}{|I(M)|} \longrightarrow m(m+q) \qquad (M \downarrow M_0). \tag{17}$$

In particular the scale curvature is strictly positive near the lower threshold also when $q > m$.

Proof. Along a convergent selected-profile subsequence define the nonnegative finite measures

$$d\mu_M(\xi) = \frac{h(\xi)|\xi|^q}{1+|\lambda_M\xi|^q}|\widehat{V}_M(\xi)|^2\, d\xi.$$

Their total masses tend to $C_{\min}$. On every compact annulus they converge to $h(\xi)|\xi|^q|\widehat{Q}(\xi)|^2\, d\xi$, whose total mass is also $C_{\min}$. Positivity therefore gives tightness away from both zero and infinity: exhaust the limiting integral by compact annuli and use total-mass convergence to control the complements.

Formula (14) becomes

$$\frac{\mathcal{K}(U_M)}{\alpha\lambda_M^{m+q}} = \frac{q}{2}\int \frac{(m+q)+(m-q)|\lambda_M\xi|^q}{(1+|\lambda_M\xi|^q)^2}\,d\mu_M(\xi).$$

The displayed multiplier is uniformly bounded and tends on compact annuli to $m+q$. Tightness proves that the right side converges to $q(m+q)C_{\min}/2$. Since $B(U_M) = \alpha\lambda_M^{m+q}\mu_M(\mathbb{R}^N)$, we have

$$\frac{\mathcal{K}(U_M)}{B(U_M)} \longrightarrow \frac{q(m+q)}{2}.$$

Combining this with (16) proves the first assertion. The energy and scale laws in (15), and $A_0(U_M) = \alpha\lambda_M^m$, give the second. Every selected subsequence has the same scalar limits, so the conclusions hold along the full family. □

There is also a fixed-scale version. Let $b_0(\xi) = a(\xi)|\xi|^q/(1+|\xi|^q)$, $k_\varepsilon = a+\varepsilon b_0$, and fix $0<\tau<1$. For the joint limit with $\alpha_\varepsilon = (1+\varepsilon\tau)^{1/r}$, suppose the selected minimizers satisfy $U_\varepsilon/\sqrt{\alpha_\varepsilon} \to Q$ strongly in H^s modulo translations. Since $(\mathcal{D}^2 - m\mathcal{D})b_0$ is an order-m multiplier,

$$\frac{\mathcal{K}(U_\varepsilon)}{\alpha_\varepsilon\varepsilon} \longrightarrow \frac{1}{2}\int (\mathcal{D}^2 - m\mathcal{D})b_0(\xi)|\widehat{Q}(\xi)|^2\,d\xi. \tag{18}$$

For $q \le m$ the limit is positive. Thus an arbitrarily weak perturbation creates a scale curvature of order ε at the finite spatial scale selected by a mass excess of the same order.

An oscillation frequency requires a further Hamiltonian spectral analysis. The conjugate phase, translation coupling, and interaction with other spectral modes enter that calculation. The exact curvature and its asymptotics above do not presume that such an isolated oscillatory eigenmode has already been established.

2.6 Which parts extend to other critical interactions

The concentration argument uses only a homogeneous energy $E_a = A/2 - \mathcal{F}$ with $E_a \ge 0$ at mass at most M_c, compactness of normalized zero-energy sequences, and a bounded quadratic correction $-D/2$. It does not use locality or a polynomial formula for $\mathcal{F}$. Theorems 2.1 and 2.4 consequently extend, with the same proof, to a mass-critical Hartree interaction once its sharp inequality, optimizer compactness, and conservative local flow have been verified. The curvature calculation needs only $\mathcal{F}(U_t) = e^{mt}\mathcal{F}(U)$, in addition to the explicit dispersion. Set stability further requires compactness of the relevant constrained minimizing sequences. This separates the common concentration mechanism from the interaction-specific analytic inputs.

3 The spectral pair created by slow translation

The preceding velocity–momentum identities determine the inertia of the charge Hessian. We now identify the pair of spectral values that emerges from the critical scaling mode, including its leading coefficient. The result assumes real-profile nondegeneracy; this hypothesis is not supplied by phase coercivity alone.

Let a be positive, elliptic, real-even and homogeneous of degree $m>1$. Consider

$$iu_t = a(D)u - |u|^{2m/N}u, \qquad a(D)U + \omega U + iv\cdot\nabla U - |U|^{2m/N}U = 0.$$

All Hilbert spaces in this section are real unless explicitly complexified. Let $Q = U_{1,0}$ be a real sharp ground wave and set

$$\Lambda Q = \frac{N}{2}Q + x\cdot\nabla Q, \qquad L_- = a(D) + 1 - |Q|^{2m/N}, \qquad L_+ = a(D) + 1 - \left(1+\frac{2m}{N}\right)|Q|^{2m/N}.$$

We impose the following hypotheses.

1. The functions $\partial_j Q$ and ΛQ have the regularity and spatial decay needed to differentiate translations and critical scaling in L^2 and in the profile equation. In particular, $L_+ \Lambda Q = -mQ$ holds in the operator sense.
2. $L_- \geq 0$, $\ker L_- = \mathbb{R}Q$, and L_-^{-1} is bounded on $Q^\perp$. Moreover $\ker L_+ = \operatorname{span}\{\partial_j Q : 1 \leq j \leq N\}$, and L_+ has a bounded inverse on that kernel's orthogonal complement.
3. There is a C^1 local profile branch $U_{\omega,v}$, closed under critical scaling up to phase and translation. Its reduced action $d(\omega, v) = E(U_{\omega,v}) + \omega M(U_{\omega,v})/2 - v \cdot P(U_{\omega,v})/2$ is C^2 and satisfies $d_\omega = M/2$ and $d_v = -P/2$. Profile differentiation is valid, and in a translation gauge
$$D_{v_j} U_{1,0} = -iL_-^{-1} \partial_j Q.$$
4. On the real space $\mathcal{H} = L^2(\mathbb{R}^N; \mathbb{C})$ the real action Hessians $\mathcal{L}_v$ are selfadjoint on a common dense domain. Their kernels are exactly
$$K_v = \operatorname{span}_{\mathbb{R}}\{iU_{1,v}, \partial_1 U_{1,v}, \ldots, \partial_N U_{1,v}\}.$$
These kernel bases vary continuously in $\mathcal{H}$. The inverse of $\mathcal{L}_v$ on $K_v^\perp$, extended by zero on K_v, exists and is norm-continuous as a bounded operator on $\mathcal{H}$ for small v.

The last hypothesis explicitly includes a uniform complementary spectral gap. Common-domain operator continuity, or suitable norm-resolvent continuity with an isolated kernel of fixed dimension, provides standard ways to verify it. It is not being inferred for every dispersion symbol.

Define the real matrix and vector
$$T_{ij} = \langle \partial_i Q, L_-^{-1} \partial_j Q \rangle, \qquad c_j = \langle \Lambda Q, L_-^{-1} \partial_j Q \rangle,$$
and the scalar
$$\mathcal{D}_Q = \langle \Lambda Q, L_-^{-1} \Lambda Q \rangle - c^T T^{-1} c. \tag{19}$$
The inverse of L_- always acts on $Q^\perp$.

Theorem 3.1 (Small-velocity spectral asymptotics)**.** *Under the four hypotheses above, T is positive definite and $\mathcal{D}_Q > 0$. If $m \neq 2$, then for every sufficiently small nonzero v, the nonzero spectrum of*
$$\mathcal{A}_v = J\mathcal{L}_v, \qquad J = -i,$$
in a fixed neighborhood of zero consists of exactly one simple pair $\lambda_\pm(v)$, satisfying
$$\lambda_\pm(v)^2 = \frac{(m-2) v^T T v}{\mathcal{D}_Q} + o(|v|^2). \tag{20}$$
For $m > 2$ this pair is real and
$$\lambda_\pm(v) = \pm \sqrt{\frac{(m-2) v^T T v}{\mathcal{D}_Q}} + o(|v|).$$
In particular, the wave is spectrally unstable. For $1 < m < 2$ the pair is purely imaginary, with
$$\lambda_\pm(v) = \pm i \sqrt{\frac{(2-m) v^T T v}{\mathcal{D}_Q}} + o(|v|).$$
The latter assertion concerns the pair near zero and does not, by itself, exclude unstable spectrum elsewhere.

Proof. Phase coercivity gives $T > 0$: a nonzero directional derivative of a nonzero L^2 function cannot vanish. Put
$$\Lambda_\perp Q = \Lambda Q - \sum_j (T^{-1} c)_j \partial_j Q.$$

Then

$$\mathcal{D}_Q = \langle \Lambda_\perp Q, L_-^{-1} \Lambda_\perp Q \rangle \geq 0.$$

Equality would imply $(x-\zeta)\cdot\nabla Q + NQ/2 = 0$ for some ζ. Thus Q would be fixed by the mass-preserving dilation group about ζ. A nonzero function homogeneous of degree $-N/2$ cannot belong to L^2: its radial mass integral is proportional to $\int_0^\infty dr/r$. Hence $\mathcal{D}_Q > 0$. The same Schur-complement formula shows that $\mathcal{D}_Q$ is unchanged if a linear combination of translations is added to ΛQ, so it is independent of the spatial center.

Write $U_v = U_{1,v}$ and $g_0 = iU_v$, $g_j = \partial_j U_v$. Their span K_v is isotropic:

$$\langle g_i, J g_j \rangle = 0. \tag{21}$$

For the phase–translation pairings this follows from integration of a mass derivative. For two translations it follows from commutation of translation charges, or directly from the Fourier representation of $\langle \partial_i U, -i\partial_j U \rangle_{\mathbb{R}}$.

Let Π_v project orthogonally onto K_v, let $\Pi_c = I - \Pi_v$, and write $\mathcal{L}_c = \mathcal{L}_v|_{K_v^\perp}$ and $J_c = \Pi_c J \Pi_c$. The operator

$$R_v(\lambda) = (\mathcal{L}_c + \lambda J_c)^{-1}$$

is extended by zero on K_v. A Neumann series and hypothesis 4 show that this inverse is analytic for $|\lambda| < \rho$, with a common $\rho > 0$ for small v. The eigenvalue problem is considered on the complexification of $\mathcal{H}$; real pairings below extend bilinearly.

The equation $J\mathcal{L}_v\psi = \lambda\psi$ is equivalent to $(\mathcal{L}_v + \lambda J)\psi = 0$. Decompose $\psi = k + w$ with $k \in K_v$ and $w \in K_v^\perp$. For $\lambda \neq 0$ the complementary equation gives

$$w = -\lambda R_v(\lambda) J k.$$

By (21), the kernel equation is therefore exactly $D(\lambda, v)z = 0$, where

$$D_{ij}(\lambda, v) = \langle g_i, J R_v(\lambda) J g_j \rangle, \qquad k = \sum_{j=0}^{N} z_j g_j. \tag{22}$$

The common factor suppressed in this equation is $-\lambda^2$. For $0 < |\lambda| < \rho$, analytic block elimination therefore preserves the eigenvalues and their algebraic multiplicities. Selfadjointness of $\mathcal{L}_c$ and skew-adjointness of J imply

$$D(-\lambda, v) = D(\lambda, v)^T. \tag{23}$$

Profile differentiation yields

$$\mathcal{L}_v \partial_\omega U = -U, \qquad \mathcal{L}_v \partial_{v_j} U = -i\partial_j U = J g_j.$$

Components of the parameter derivatives in K_v do not contribute, by (21). Consequently

$$D(0, v) = C d''(1, v) C, \qquad C = \mathrm{diag}(1, -I_N).$$

The lower-right block is $D_{vv}(0, v) = -B(1, v)/2$, where $B = D_v P$. The first derivative of the branch gives

$$P(U_v) = 2Tv + o(|v|), \qquad B(1, v) = 2T + o(1).$$

Thus the lower-right block stays invertible for small λ, v. Take its scalar Schur complement $f(\lambda, v)$. Identity (23) makes f even in λ. It is consequently analytic in $z = \lambda^2$ in a common disk, with coefficients continuous in v.

The exact scaling identity for the reduced action gives

$$\begin{aligned} f(0, v) &= \frac{P^T B^{-1} P - (m-1) v \cdot P}{2m^2} \\ &= \frac{2-m}{m^2} v^T T v + o(|v|^2). \end{aligned} \tag{24}$$

For completeness, the identity follows by differentiating the scaling laws for the charges. With $H = d''$ and $D_* = (m\omega, (m-1)v)^T$, they give $HD_* = (0, -P/2)^T$ and $H_{vv} = -B/2$. Eliminating the velocity block shows that its scalar Schur complement is $[P^T B^{-1} P - (m-1)v \cdot P]/(2m^2\omega^2)$. This uses no differentiated second-order remainder for the profile.

It remains to compute the coefficient of λ^2 at $v = 0$. Let R_0 be the L^2-orthogonal projection of $\Lambda Q/m$ off the translation kernel, and put $Z_j = L_-^{-1}\partial_j Q$. At $v = 0$ the two real blocks give

$$\mathcal{L}_0 = \operatorname{diag}(L_+, L_-), \qquad J(h, b) = (b, -h),$$

$$\mathcal{L}_c^{-1} J g_0 = (-R_0, 0), \qquad \mathcal{L}_c^{-1} J g_j = (0, -Z_j).$$

Expanding the exact resolvent in (22), and writing $b_j = \langle R_0, Z_j \rangle$, gives

$$\begin{aligned} D_{00}(\lambda, 0) &= \lambda^2 \langle R_0, L_-^{-1} R_0 \rangle + O(\lambda^4), \\ D_{0v}(\lambda, 0) &= \lambda b^T + O(\lambda^2), \\ D_{v0}(\lambda, 0) &= -\lambda b + O(\lambda^2), \\ D_{vv}(\lambda, 0) &= -T + O(\lambda). \end{aligned}$$

The scalar Schur complement is even. Its coefficient is therefore

$$\partial_z f(0, 0) = \langle R_0, L_-^{-1} R_0 \rangle - b^T T^{-1} b = \frac{\mathcal{D}_Q}{m^2} > 0.$$

The final equality uses invariance of the Schur expression under addition of translations to the scaling derivative. Analyticity with a common radius and continuity in v now give

$$f(\lambda, v) = \frac{2-m}{m^2} v^T T v + o(|v|^2) + \lambda^2 \left(\frac{\mathcal{D}_Q}{m^2} + o(1) \right) + O(\lambda^4),$$

where the last remainder is uniform for small v.

The analytic scalar equation, regarded as an equation in $z = \lambda^2$, has one simple real root near zero, by $\partial_z f(0, 0) > 0$ and uniform analytic inversion. That root obeys (20). Its sign is the sign of $m - 2$, since $T > 0$, and it is nonzero for nonzero sufficiently small v. Its two square roots are simple because $\partial_\lambda f = 2\lambda \partial_z f \neq 0$ there. The invertible complementary and velocity blocks exclude further nonzero spectrum in a sufficiently small fixed neighborhood. □

Remark 3.2 (Center, frequency, and linear time scale). If $Q(-x) = Q(x)$, then $c = 0$ by parity. Without this symmetry, omitting $c^T T^{-1} c$ gives an incorrect coefficient in general. Critical scaling also gives the exact spectral covariance

$$\lambda(\omega, v) = \omega \lambda\big(1, v\omega^{-(m-1)/m}\big).$$

Thus for $m > 2$ and fixed $\omega > 0$,

$$\lambda_+(\omega, v) = \omega^{1/m} \sqrt{\frac{(m-2) v^T T v}{\mathcal{D}_Q}} + o(|v|).$$

The reciprocal growth rate is of order $|v|^{-1}$. This is a statement about linear exponential growth, not an established nonlinear escape time. When $m = 2$, the displayed leading coefficient vanishes; a general homogeneous degree-two symbol need not be a quadratic form, so an exact Galilean degeneracy is not asserted.

Corollary 3.3 (Critical Hartree interactions)**.** *Theorem 3.1 holds, with the same coefficient, for the critical Hartree flow*

$$i u_t = a(D) u - (K * |u|^2) u, \qquad K(x) = |x|^{-m}, \quad 1 < m < N,$$

whenever its real ground wave and local branch satisfy the same four hypotheses with

$$L_- = a(D) + 1 - (K * Q^2), \qquad L_+ h = L_- h - 2Q\big(K * (Qh)\big).$$

It also holds for a positive-type anisotropic interaction of the same homogeneity under those operator and regularity assumptions.

Proof. The proof above uses the nonlinearity through phase and translation invariance, conjugation symmetry, critical scaling, and the stated real-block operators. These properties hold for the Hartree energy. In particular $L_+ \Lambda Q = -mQ$, the parameter equations and the charge-scaling identities are unchanged. No step in the resolvent reduction uses locality of L_+. Thus the proof applies verbatim. □

Remark 3.4 (Relation to established instability theory). The general mechanism of a spectral pair leaving zero at charge Hessian degeneracy belongs to established Hamiltonian bifurcation theory. Here the additional calculation is its coefficient for mass-critical translation branches, including the coupling to the translation modes. The theorem proves spectral instability for $m > 2$ under its explicit assumptions. Nonlinear orbital instability requires a sufficiently differentiable local flow and the associated linear evolution bounds or unstable-manifold construction in a specified phase space; it is not inferred here for every admissible symbol and dimension. Likewise, real-profile nondegeneracy is not proved by this section.

4 Least traveling mass and degenerate neutral modes

Let $m > 1$, $s = m/2$, and let a be a real-even, positive elliptic m-homogeneous symbol with the regularity needed for the energy-space profile equation and its Pohozaev identities. On $X = H^s(\mathbb{R}^N; \mathbb{C})$, considered as a real space, put

$$E(U) = \frac{1}{2}\int a(\xi)|\widehat{U}(\xi)|^2\, d\xi - \frac{1}{p}\int |U|^p, \qquad p = 2 + \frac{2m}{N},$$
$$M(U) = \|U\|_2^2, \qquad P(U) = \operatorname{Im}\int \overline{U}\nabla U.$$

An exact profile satisfies

$$F(U, \omega, v) := a(D)U + \omega U + iv \cdot \nabla U - |U|^{p-2}U = 0. \tag{25}$$

The action, its real Hessian, and the Hamiltonian structure are

$$S_{\omega,v} = E + \omega M/2 - v \cdot P/2, \qquad L = S''_{\omega,v}(U), \qquad J = -i.$$

Assume the negative-energy profile set is nonempty and define

$$M_{\mathrm{tr}} = \inf\{M(U) : F(U, \omega, v) = 0 \text{ for some } (\omega, v),\ E(U) < 0\}.$$

The identity $v \cdot P = 2mE$ is understood with this normalization. All parameter families below are defined on open parameter sets; they are C^1 into X, differentiation of the profile equation is valid in X^*, and their reduced action is C^2.

4.1 Mass extrema and the reduced action

For a full profile family U_θ, $\theta = (\omega, v)$, write

$$d(\theta) = S_\theta(U_\theta), \qquad H = d''(\theta), \qquad B = D_v P(U_\theta), \qquad e_\omega = (1, 0, \dots, 0)^T.$$

Stationarity implies the exact identities

$$\nabla d = (M/2, -P/2)^T, \qquad E = d - \theta \cdot \nabla d, \qquad \nabla E = -H\theta, \qquad \nabla M = 2He_\omega. \tag{26}$$

In particular, $H\theta = 0$ is a null-vector identity and does not mean that the entire matrix H vanishes.

If the charges of the family obey critical scaling covariance, differentiating that covariance and using the symmetry of H gives

$$m\omega M_\omega + (m-1)v \cdot M_v = 0, \qquad m\omega P_\omega + (m-1)Bv = P, \qquad M_v = -P_\omega. \tag{27}$$

This covariance holds for a family locally unique modulo phase and translation, because critical dilation produces another exact profile with parameters $(\lambda^m\omega, \lambda^{m-1}v)$.

Theorem 4.1 (An attained negative-energy endpoint)**.** *Suppose a member of a full profile family satisfies $E(U) < 0$ and $M(U) = M_{\mathrm{tr}}$. Then*

$$M_\omega = 0, \qquad M_v = 0, \qquad P_\omega = 0, \qquad He_\omega = 0. \tag{28}$$

If the family has critical scaling covariance, then additionally

$$(m-1)Bv = P, \qquad v^T Bv = \frac{v \cdot P}{m-1} < 0. \tag{29}$$

Thus the charge Hessian is singular, whether or not B is invertible.

Proof. Negative energy persists in a parameter neighborhood. By definition of M_{tr}, mass has a local minimum there at the profile under consideration. Its first derivatives vanish. Equations (26) and (27) give the conclusions, and the strict sign follows from $v \cdot P = 2mE < 0$. □

The supplied-family hypothesis can be replaced by the following standard local profile assumptions. Suppose F is C^1 as a map into X^*, $L : X \to X^*$ is selfadjoint Fredholm, and

$$\ker L = K := \mathrm{span}_{\mathbb{R}}\{iU, \partial_1 U, \ldots, \partial_N U\}. \tag{30}$$

Assume the profile and projected solutions have enough regularity for these generators to belong to X and for the symmetry Noether pairings below to hold. These are local regularity assumptions, not an assertion that translations are differentiable at every point of X.

Lemma 4.2 (Local profile family from a symmetry slice)**.** *Under these assumptions, the profile has a full C^1 local parameter family, unique up to phase and translation, and its charges have critical scaling covariance.*

Proof. The generators in (30) are independent: a real dependence implies a constant-coefficient first-order equation for U, forcing its L^2 Fourier transform onto an affine hyperplane, or forcing $U = 0$. Take the fixed L^2-orthogonal slice to K in X. By selfadjoint Fredholmness, the restriction of L is an isomorphism from that slice onto the annihilator of K in X^*. Apply the implicit-function theorem to the projected equation. Its residual is a linear combination of the original generators, identified with elements of X^*. Gauge and translation invariance of $S_{\omega,v}$ give

$$\langle F(U_{\mathrm{new}}, \omega, v), iU_{\mathrm{new}}\rangle_{\mathbb{R}} = 0, \qquad \langle F(U_{\mathrm{new}}, \omega, v), \partial_j U_{\mathrm{new}}\rangle_{\mathbb{R}} = 0.$$

The pairing matrix of the original and new generators is invertible near U. Thus all residual coefficients vanish. Local slice uniqueness gives uniqueness modulo symmetries. Applying that uniqueness to the critically dilated profiles proves covariance of their charges. □

For example, a positive free spectral gap $\omega > \sup_\xi(v \cdot \xi - a(\xi))$, together with compactness of the nonlinear potential perturbation, supplies the Fredholm hypothesis. The kernel and local regularity assumptions must still be verified.

4.2 A degenerate charge response yields secular linear growth

Theorem 4.3 (An extended neutral Jordan chain)**.** *Assume* (30) *and the regularity of Lemma 4.2. If* $0 \neq q = (q_\omega, q_v) \in \ker H$*, set*

$$Z = D_\theta U[q], \qquad g = q_\omega iU - q_v \cdot \nabla U.$$

There exists $W \in X$ *such that*

$$LW = iZ, \qquad JLW = Z, \qquad JLZ = g, \qquad JLg = 0. \tag{31}$$

The chain has length three. If JL *generates the linearized evolution on a phase space containing these vectors in its generator domain, then*

$$e^{tJL}W = W + tZ + \frac{t^2}{2}g. \tag{32}$$

In the quotient by the phase and translation tangent space K*, this solution grows at least linearly as* $t \to +\infty$*.*

Proof. Parameter differentiation gives $LU_\omega = -U$ and $LU_{v_j} = -i\partial_j U$, so $JLZ = g$. Solvability of $LW = iZ$ is equivalent to iZ annihilating K. Its pairings with iU and $\partial_j U$ are, respectively, $DM(U)[Z]/2$ and $DP_j(U)[Z]/2$. These vanish precisely because $Hq = 0$. Selfadjoint Fredholm solvability gives W. Independence of the symmetry generators implies $g \neq 0$; hence $Z \notin K$. The finite Jordan chain proves (32). In X/K its value is $[W] + t[Z]$, whose norm is at least $t\|[Z]\| - \|[W]\|$. □

At the attained endpoint of Theorem 4.1, one may take $q = e_\omega$. The conclusion is secular growth of the linearized equation modulo symmetries. It asserts neither a positive-real-part eigenvalue nor nonlinear orbital instability.

4.3 Zero-energy components of a threshold limit

The preceding compactification result for M_{tr} says that a mass-minimizing negative-energy sequence, normalized to kinetic energy one, either converges strongly modulo translation to a negative-energy profile or splits strongly into finitely many zero-energy profiles with common parameters. In the splitting alternative,

$$\sum_j M(V_j) = M_{\mathrm{tr}}, \qquad \omega_* = (m/N)/M_{\mathrm{tr}} > 0.$$

The following statements add restrictions to those alternatives.

Theorem 4.4 (Degeneracy of each genuinely split component)**.** *Let* V *be a zero-energy exact profile with* $M(V) < M_{\mathrm{tr}}$ *belonging to a full profile family. Then*

$$\nabla_\theta E = 0, \qquad H\theta = 0. \tag{33}$$

For a scaling-covariant family with $\omega \neq 0$*, one also has*

$$Bv = -P, \qquad M_v = -P/\omega, \qquad M_\omega = 0. \tag{34}$$

Consequently every component in a split threshold limit with at least two nonzero components has singular charge Hessian whenever it has a regular profile slice as above; it then has the neutral chain of Theorem 4.3.

Proof. All sufficiently nearby members of the family still have mass less than M_{tr}. Their energy is nonnegative by definition of that threshold. Thus E has a local minimum at V and (26) gives (33). Differentiating $E = v \cdot P/(2m)$ gives $Bv = -P$. Substitution into (27) gives $M_v = -P/\omega$, and its first identity, together with $v \cdot P = 0$, gives $M_\omega = 0$. In a genuine finite split every component has mass strictly below the sum, and $\theta_* \neq 0$ because $\omega_* > 0$. □

If the family is rotationally equivariant for an isotropic equation, $P = p(c)v/c$ for $c = |v| > 0$. At a component in Theorem 4.4, zero energy gives $p(c) = 0$ and $Bv = -P$ gives $p'(c) = 0$. Hence $B = 0$; when $\omega \neq 0$, (34) shows that the *entire* matrix H vanishes. Isotropy of the equation alone does not assert rotational equivariance of every profile family.

Proposition 4.5 (A moving zero-energy endpoint)**.** *Suppose $E(U) = 0$, $M(U) \leq M_{\mathrm{tr}}$, $v \neq 0$, and U belongs to a full profile family. Then H is singular.*

Proof. If H were nonsingular, the map $\theta \mapsto (M, P)$ would be a local C^1 diffeomorphism. Along exact profiles, stationarity gives

$$dE = -\frac{\omega}{2}\, dM + \frac{v}{2} \cdot dP.$$

Choose $dM < 0$ and then choose dP with a sufficiently negative component along v that $dE < 0$. A small positive perturbation in this charge direction has both negative energy and mass below M_{tr}, a contradiction. □

Remark 4.6 (The standing-wave exception)*.* At a normalized zero-energy endpoint $\omega > 0$. If $v = 0$ and $P = 0$, scaling gives $H(m\omega, 0)^T = (0, -P/2)^T = 0$, so H is again singular. The remaining case $v = 0$, $P \neq 0$ is not excluded. The present argument does not prove that every complex stationary profile is real up to a constant phase, or that every such profile has zero momentum. Thus the correct statement for a single regular zero-energy endpoint retains the possibility of a standing profile with nonzero momentum.

4.4 The exact turning-point slope in an isotropic branch

Proposition 4.7. *Suppose an isotropic rotationally equivariant branch, at fixed $\omega > 0$, reaches an attained negative-energy mass threshold at speed $c_* > 0$. Write $P = p(c)v/c$, and assume the differentiability used below. Then*

$$p(c_*) < 0, \qquad p'(c_*) = \frac{p(c_*)}{(m-1)c_*} < 0.$$

The matrix B is negative definite at c_*, and H has N positive eigenvalues and a one-dimensional kernel. If the mass minimum satisfies $M''(c_*) > 0$, define

$$\kappa(c) = \frac{p(c)^2}{p'(c)} - (m-1)cp(c).$$

Then

$$\kappa(c_*) = 0, \qquad \kappa'(c_*) = -m\omega(m-1)c_* M''(c_*) < 0. \tag{35}$$

Near c_* the positive index of H drops from $N+1$ to N as c increases through c_*.

Proof. Use (29); the transverse eigenvalues of B are $p(c_*)/c_* < 0$, and the longitudinal one is $p'(c_*) < 0$. At the endpoint the first row and column of H vanish and $H_{vv} = -B/2$, proving its stated inertia. Stationarity of S and $E = cp/(2m)$ give

$$m\omega M' = (m-1)cp' - p.$$

Differentiate this identity once. Direct differentiation of κ at c_*, using $(m-1)c_*p' = p$, gives

$$\kappa' = (2-m)p - (m-1)^2 c_*^2 p'' = -m\omega(m-1)c_* M''.$$

Finally the Schur complement of the v block in H equals $\kappa/(2m^2\omega^2)$, obtained from (27) and $H_{vv} = -B/2$. The v block remains positive definite nearby and κ changes from positive to negative. □

The relation between conserved-charge degeneracy and extended neutral Jordan chains belongs to classical Hamiltonian solitary-wave theory, including the work of Grillakis–Shatah–Strauss and Comech–Pelinovsky. The conclusions here specify what that mechanism forces at the least-negative-energy traveling-mass threshold and in its finite splitting. The turning formula alone is not an eigenvalue bifurcation theorem; that requires the additional spectral hypotheses and resolvent expansion stated separately.

5 Polarization rigidity with arbitrarily many internal components

The density geometry has a consequence which is stronger than a componentwise extension: near criticality, a common-frequency traveling wave cannot acquire independent spatial profiles in its internal components. The number of components may be unbounded.

5.1 Class of systems and main statements

Let $\mathcal{H}$ be any nonzero separable complex Hilbert space. Finite-dimensional $\mathbb{C}^d$ and ℓ^2 are both allowed. Fix $N \geq 1$, $m > 1$, $s = m/2$, and a real-even, positive elliptic, m-homogeneous symbol a, of class C^{N+2} away from zero. For $U \in H^s(\mathbb{R}^N; \mathcal{H})$ put

$$\rho_U = \|U\|_{\mathcal{H}}^2, \quad M(U) = \int \rho_U, \quad A(U) = \int a(\xi)\|\widehat{U}(\xi)\|_{\mathcal{H}}^2 \, d\xi.$$

We use either of the following interactions:

$$\begin{aligned} &\kappa = 1 + m/N, && \mathcal{F}(U) = \int \rho_U^\kappa, && V_\rho = \rho^{\kappa-1}; \\ &\kappa = 2, && \mathcal{F}(U) = \mathcal{D}(\rho_U, \rho_U), && V_\rho = K * \rho. \end{aligned}$$

In the second case $1 < m < N$, the real even kernel K is homogeneous of degree $-m$, its interaction form $\mathcal{D}$ is positive definite, and

$$|K(x)| + |x||\nabla K(x)| \leq C|x|^{-m}.$$

As before, positivity of the Fourier transform of K almost everywhere suffices for positive definiteness. Pointwise positivity is not needed. The scalar sharp inequality defines M_c by

$$\mathcal{F}(u) \leq \kappa \left(\frac{M(u)}{M_c}\right)^{\kappa-1} A(u). \tag{36}$$

Allow a common scalar addition $b \geq 0$, real and even, such that

$$\xi \cdot \nabla b \leq mb, \qquad |\partial^\beta b(\xi)| \leq C_\beta |\xi|^{-|\beta|}(1 + |\xi|^m), \quad |\beta| \leq N+2. \tag{37}$$

The constants need not be bounded over the class. Write

$$B_b(U) = \int b(\xi)\|\widehat{U}(\xi)\|^2 \, d\xi, \qquad E_b(U) = \frac{A(U) + B_b(U)}{2} - \frac{\mathcal{F}(U)}{2\kappa}.$$

A common-frequency traveling profile satisfies

$$[a(D) + b(D) + \omega + iv \cdot \nabla]U - V_{\rho_U} U = 0. \tag{38}$$

All components have the same a, b, ω, v. This is the profile of $u(t,x) = e^{i\omega t}U(x - vt)$.

Theorem 5.1 (Sharp interpolation and dimension-independent compactness)**.** *Inequality* (36) *has the same sharp constant for $\mathcal{H}$-valued functions. Every optimizer has the form*

$$U(x) = \zeta Q(x), \qquad \|\zeta\|_{\mathcal{H}} = 1,$$

where Q is a real scalar optimizer, up to a constant phase. If $M(U_n) \to M_c$, $A(U_n) \to 1$, and $\mathcal{F}(U_n) \to \kappa$, then the sequence is strongly compact in H^s after translations and internal unitaries. This remains true when the internal dimensions vary: embed all spaces in ℓ^2 before applying the unitaries.

Theorem 5.2 (Nonpositive-energy rigidity, uniform in internal dimension)**.** *There is $\delta > 0$, depending only on N, m, a and the scalar interaction, such that every nonzero weak profile* (38) *satisfying*

$$M(U) < M_c + \delta, \qquad E_b(U) \le 0$$

has

$$E_b(U) = 0, \qquad v = 0, \qquad D_b(U) = 0, \qquad U = \zeta q,$$

where

$$D_b(U) = \int [mb - \xi \cdot \nabla b] \|\widehat{U}\|^2$$

and q is a real scalar stationary profile for $a+b$ and ω. Necessarily $M(U) \ge M_c$. In particular, every negative-energy profile has mass at least $M_c + \delta$, uniformly over $\mathcal{H}$, the cone (37)*, all frequencies and all velocities. If $mb - \xi \cdot \nabla b > 0$ almost everywhere away from zero, no nonzero nonpositive-energy profile lies in this mass window.*

Corollary 5.3 (Exact polarization near criticality)**.** *For $b = 0$, there is $\eta > 0$, independent of $\mathcal{H}$, such that every nonzero weak profile satisfying*

$$\left| \frac{M(U)}{M_c} - 1 \right| + \frac{|E_0(U)|}{A(U)} < \eta$$

is exactly polarized: $U = \zeta q$ for a scalar traveling profile q. Positive energy is allowed in this statement.

5.2 Sharp interpolation and complete polarization

The local interaction is the κ-th power of the L^κ density norm. The Hartree interaction is the square of its positive definite Hilbert density norm. Thus both have the form $\mathcal{F}(U) = \|\rho_U\|_{\mathcal{B}}^\kappa$ for a strictly convex real density space $\mathcal{B}$. In an internal orthonormal basis $U = (u_j)$, the triangle inequality, scalar sharp interpolation, and sequence Hölder give

$$\begin{aligned}
\mathcal{F}(U)^{1/\kappa} &\le \sum_j \mathcal{F}(u_j)^{1/\kappa} \\
&\le \kappa^{1/\kappa} M_c^{-(\kappa-1)/\kappa} \sum_j A(u_j)^{1/\kappa} M(u_j)^{(\kappa-1)/\kappa} \\
&\le \kappa^{1/\kappa} M_c^{-(\kappa-1)/\kappa} A(U)^{1/\kappa} M(U)^{(\kappa-1)/\kappa}.
\end{aligned}$$

Finite partial sums converge in the density norm because the same bound applied to their tails tends to zero. A scalar optimizer placed in one component proves sharpness.

Split the components further into real scalar functions f_j. Equality forces every active f_j to be a scalar optimizer, all f_j^2 to be proportional, and all kinetic-to-mass ratios to agree. The same conclusions hold after any real orthogonal rotation of a pair f_i, f_j. If f_j/f_i took both signs of a nonzero constant, the squared sum and difference could not be proportional; when the constant has magnitude one their nonzero supports would be disjoint. Thus each f_j is one real constant multiple of a fixed scalar function. This proves complete polarization, including for countably many components.

5.3 Compactness modulo internal unitaries

Ordinary local compactness does not hold directly in an infinite internal dimension: a fixed spatial shape can move through mutually orthogonal internal vectors. The following argument removes precisely that loss of compactness.

Lemma 5.4 (Internal-unitary local compactness)**.** *Let $X = H^s(\mathbb{R}^N;\mathbb{C})$ with $s > 0$. From every bounded sequence in $X\widehat{\otimes}\ell^2$, one can extract a subsequence and internal unitaries W_n such that W_nU_n converges weakly in $X\widehat{\otimes}\ell^2$ and strongly in $L^2_{\mathrm{loc}}(\mathbb{R}^N;\ell^2)$.*

Proof. Fix an orthonormal basis (ϕ_k) of the scalar Hilbert space X. Write its internal coefficient vectors as $c_{nk} = \langle \phi_k, U_n\rangle_X \in \ell^2$. Gram–Schmidt the first n vectors and extend the resulting finite orthonormal list to an internal orthonormal basis. A unitary W_n can therefore be chosen so that

$$W_n c_{nk} \in \operatorname{span}\{e_1,\dots,e_k\}, \qquad k \le n.$$

For each fixed k, these vectors have a convergent subsequence in a fixed finite-dimensional space. Diagonal extraction gives strong convergence of every coefficient vector. Weak compactness gives a weak limit of W_nU_n, with exactly those coefficients.

For $\chi \in C_c^\infty$, the scalar multiplication map $K_\chi : X \to L^2$, $f \mapsto \chi f$, is compact. If P_L is the scalar projection onto the first L basis vectors, then

$$\|K_\chi(I - P_L)\|_{X\to L^2} \longrightarrow 0.$$

Tensoring with the identity on ℓ^2 preserves this norm. The transformed P_LU_n converge strongly because there are only L coefficient vectors; after multiplication by χ, the tails are uniformly small. This proves local strong convergence. □

Compactness assertion of Theorem 5.1. The localized Sobolev inequality has constants independent of internal dimension: use the Hilbert-valued Bessel-potential representation and its norm triangle inequality, then spatial localization and the sum of component quadratic norms. Thus local mass vanishing for an H^s-bounded sequence implies vanishing in $L^p(\mathcal{H})$ for $2 < p < 2N/(N-2s)$, or every finite $p > 2$ if $2s \ge N$. In the local case take $p = 2\kappa$; in the Hartree case use $p = 4N/(2N-m)$ and Hardy–Littlewood–Sobolev. Hence $\mathcal{F}(U_n) \to \kappa > 0$ excludes local mass vanishing.

Translate a nonzero amount of local mass into a fixed ball and use Lemma 5.4. We obtain a nonzero weak limit U, strong local L^2 convergence, and by interpolation strong local convergence in the interaction exponent. Put $R_n = U_n - U$. The quadratic quantities split. For local powers the Brezis–Lieb argument for $\|U\|_{\mathcal{H}}^{2\kappa}$ gives

$$\mathcal{F}(U_n) = \mathcal{F}(U) + \mathcal{F}(R_n) + o(1).$$

For Hartree, the term $\operatorname{Re}\langle U, R_n\rangle$ tends to zero in $L^{p/2}$ by local convergence and the small L^p tail of U. Also $\mathcal{D}(\rho_U, \rho_{R_n}) \to 0$: approximate U in $L^p(\mathcal{H})$ by a compactly supported smooth finite-dimensional field. Its potential is bounded and decays at infinity, while R_n tends locally to zero in L^2. HLS controls the approximation error. This proves the same splitting for Hartree.

Let $M_1 > 0, M_2$ and A_1, A_2 be the limiting contributions. Then $M_1 + M_2 = M_c$, $A_1 + A_2 = 1$, and sharp interpolation yields

$$M_c^{\kappa-1} \le M_1^{\kappa-1}A_1 + M_2^{\kappa-1}A_2.$$

Both masses cannot be positive, since each would be strictly below M_c. Thus $M_2 = 0$, and the inequality then forces $A_2 = 0$. The convergence is strong in H^s. The limit is an optimizer, already proved to be completely polarized. □

5.4 Phase coercivity and the internal symmetry kernel

For completeness, the scalar phase fact used below follows directly from the density norm. Normalize a real optimizer by $M(Q) = M_c$, $A(Q) = 1$, and put $\omega_* = (\kappa - 1)/M_c$. The form

$$L_Q = a(D) + \omega_* - V_{Q^2}$$

is nonnegative: differentiate the sharp inequality at $Q + i\sqrt{t}\,h$. If h belongs to its kernel, write $x = A(h)$ and $y = M(h)/M_c$. The supporting-functional inequality for the density norm, sharp interpolation for h, and weighted arithmetic–geometric mean give

$$x + (\kappa - 1)y \le \kappa x^{1/\kappa} y^{(\kappa-1)/\kappa} \le x + (\kappa - 1)y.$$

Equality implies $x = y$ and $h^2 = yQ^2$ by strict convexity. The kernel of a nonnegative form is linear; apply the same conclusion to $h + Q$ to get $h = cQ$. The potential is compact as a form $H^s \to H^{-s}$, by local Rellich and Hölder in the local case, or the compact-potential approximation and HLS in the Hartree case. Consequently

$$\langle h, L_Q h\rangle \ge c_Q \|h\|_{H^s}^2, \qquad \langle h, Q\rangle_{L^2} = 0. \tag{39}$$

This holds for complex h by real–imaginary splitting, and it can be summed over arbitrarily many components with the same constant.

At Qe_1, write a perturbation as $(h + ik)e_1 + Z$ with $Z \perp e_1$. The Hessian of $E + \omega_* M/2$ is

$$\langle h, L_+ h\rangle + \langle k, L_Q k\rangle + \sum_{j\ge 2} \big(\langle \operatorname{Re} Z_j, L_Q \operatorname{Re} Z_j\rangle + \langle \operatorname{Im} Z_j, L_Q \operatorname{Im} Z_j\rangle\big).$$

Its kernel in the phase and internal transverse sector is exactly

$$\{Q\eta :\ \eta \in \mathcal{H}, \operatorname{Re}\langle \eta, e_1\rangle = 0\}.$$

For $\mathcal{H} = \mathbb{C}^d$, this is the $(2d-1)$-dimensional real tangent space to the internal unitary orbit. Any additional full Hessian kernel lies in the scalar real sector L_+; no assumption on that sector is made.

5.5 Quantitative polarization without real-sector nondegeneracy

The preceding compactness and phase coercivity also give a quantitative statement for arbitrary fields, not just stationary or traveling ones. Define the scale-invariant interpolation deficit

$$\mathfrak{d}(U) = 1 - \frac{\mathcal{F}(U)}{\kappa (M(U)/M_c)^{\kappa-1} A(U)}$$

and the squared distance to a completely polarized field by

$$\mathfrak{p}(U) = \inf_{\substack{\|\zeta\|_{\mathcal{H}}=1 \\ q\in H^s}} \left\{ \frac{M(U - \zeta q)}{M(U)} + \frac{A(U - \zeta q)}{A(U)} \right\}.$$

Both are defined for every nonzero $U \in H^s(\mathcal{H})$.

Theorem 5.5 (Dimension-independent quantitative polarization)**.** *There is $C < \infty$, depending only on the fixed scalar dispersion and interaction, such that*

$$\mathfrak{p}(U) \le C\mathfrak{d}(U) \tag{40}$$

for every nonzero Hilbert-valued field U and every internal dimension. No nondegeneracy or uniqueness of the scalar real-sector ground state is required.

Proof. Amplitude and spatial scaling preserve both quantities, so impose $M(U) = M_c$, $A(U) = 1$. Write $\|U\|_X^2 = M(U)/M_c + A(U)$. Since $\mathfrak{p}(U) \le 2$, failure of a uniform constant would produce U_n with $\mathfrak{d}(U_n) \to 0$ and $\mathfrak{d}(U_n)/\mathfrak{p}(U_n) \to 0$. After the symmetries of Theorem 5.1, $U_n \to Qe_1$ strongly in H^s. Align $\int QU_n$ with the positive real first internal direction and write $U_n = (u_n, Z_n)$. Each component of Z_n is complex L^2-orthogonal to Q, and $u_n \to Q$, $Z_n \to 0$ in H^s.

The interaction expansion in these transverse directions is

$$\mathcal{F}(u_n, Z_n) = \mathcal{F}(u_n) + \kappa\langle V_{|u_n|^2}, \|Z_n\|^2\rangle + o(\|Z_n\|_X^2). \tag{41}$$

For local powers, set $a = |u_n|^2$ and $b = \|Z_n\|^2$ pointwise. When $1 < \kappa \le 2$, the nonnegative Taylor remainder of $(a+b)^\kappa$ after its linear term in b is at most $C_\kappa b^\kappa$. When $\kappa > 2$, it is at most $C_\kappa(a^{\kappa-2}b^2 + b^\kappa)$. Hölder and the dimension-independent $L^{2\kappa}$ Sobolev bound make these respectively $O(\|Z_n\|_X^{2\kappa})$ and $O(\|Z_n\|_X^4 + \|Z_n\|_X^{2\kappa})$, since u_n stays bounded. Both are $o(\|Z_n\|_X^2)$. For Hartree the remainder is exactly $\mathcal{D}(\|Z_n\|^2, \|Z_n\|^2) = O(\|Z_n\|_X^4)$.

Put $m_n = M(Z_n)$ and $a_n = A(Z_n)$. Scalar sharp interpolation gives

$$\mathcal{F}(u_n) \le \kappa(1 - m_n/M_c)^{\kappa-1}(1 - a_n).$$

Combining its expansion with (41) and potential form convergence yields

$$\begin{aligned}
\kappa - \mathcal{F}(U_n) &\ge \kappa\left[a_n + \frac{\kappa-1}{M_c}m_n - \langle V_{Q^2}, \|Z_n\|^2\rangle\right] - o(\|Z_n\|_X^2) \\
&= \kappa\sum_j \langle Z_{nj}, L_Q Z_{nj}\rangle - o(\|Z_n\|_X^2) \\
&\ge c\|Z_n\|_X^2.
\end{aligned}$$

The constant is independent of component number. Taking the candidate $e_1 u_n$ in the distance gives $\mathfrak{p}(U_n) \le \|Z_n\|_X^2$. As $\mathfrak{d}(U_n) = (\kappa - \mathcal{F}(U_n))/\kappa$, this contradicts the assumed ratio and proves the theorem. □

The theorem controls internal polarization rather than the full scalar shape. It therefore remains valid even when scalar optimizer uniqueness or real-sector quantitative stability is unavailable.

5.6 Profile identities and nonpositive-energy rigidity

Weak profiles have $U \in L^\infty(\mathcal{H}) \cap H^{m+1}(\mathcal{H})$. Here the regularity argument is unaffected by the internal dimension. Choose a shifted scalar resolvent with symbol comparable to $1 + |\xi|^m$. Its $L^p(\mathcal{H}) \to H^{m,p}(\mathcal{H})$ bounds hold by Hilbert-valued Mihlin estimates. In the local case, for $N > m$, the reciprocal Sobolev exponents improve under $\beta \mapsto (1 + 2m/N)\beta - m/N$, starting at $\beta = (1 - m/N)/2$; weakening intermediate exponents avoids endpoints and gives boundedness in finitely many steps. The cases $N \le m$ start at arbitrarily large finite exponent or at boundedness. For Hartree, the near kernel is in L^j for $j < N/m$ and the far kernel is bounded. Since $\rho_U \in L^{N/(N-m)}$, the near potential belongs to arbitrarily large finite L^t. Resolvent iteration with $t > N/m$ again gives boundedness. Bounded nonlinearity coefficients then give $H^m \subset H^1$; differentiating the nonlinearity once gives H^{m+1}. The derivative of the local vector map is bounded on bounded sets. In the Hartree case use $\nabla\rho_U \in L^1 \cap L^2$ and the near/far split.

Let $P(U) = \int \xi\|\widehat{U}\|^2$, and abbreviate A, B, M, E, D_b by A, B, M, E, D. Cutoff dilation and testing by U give

$$v \cdot P = 2mE - D, \qquad \omega M = (\kappa - 1)(A + B) + (2m - 2\kappa)E - D. \tag{42}$$

These require no spatial moment. For the kinetic cutoff use $f_R = x\chi(x/R)$ and the uniformly bounded commutator with $f_R \cdot \nabla + \operatorname{div} f_R/2$. The symbol difference estimate

$$|(a+b)(\xi) - (a+b)(\eta)||\xi + \eta| \le C|\xi - \eta|[1 + (|\xi| + |\eta|)^m]$$

gives a uniform $H^s \to H^{-s}$ form bound after dividing by the two Sobolev weights. Schwartz approximation gives the limit $\int \xi \cdot \nabla(a+b)\|\widehat{U}\|^2$. The local nonlinear term is integrated by parts. The Hartree term is symmetrized into

$$-\frac{1}{4}\iint \rho_U(x)\rho_U(y)[f_R(x)-f_R(y)]\cdot\nabla K(x-y)\,dx\,dy,$$

which tends to $m\mathcal{F}(U)/4$ by HLS and the uniform Lipschitz bound on f_R. Finite internal projection followed by convergence justifies all component sums.

Write $U = X + iY$ in an internal basis and set $\widetilde{\omega} = \omega + D/M$. The imaginary equation and $P(U) = -2\int\langle Y, \nabla X\rangle$ give

$$mE = \langle Y, [a(D) + \widetilde{\omega} - V_{\rho_U}]Y\rangle + B_b(Y) + D\left(\frac{1}{2} - \frac{M(Y)}{M}\right), \tag{43}$$

$$\widetilde{\omega}M = (\kappa-1)(A+B) + (2m-2\kappa)E. \tag{44}$$

Proof of Theorem 5.2. Sharp interpolation excludes $M < M_c$ for $E \le 0$. Suppose a sequence violating the stated rigidity had $M_n \to M_c$. Critical scaling preserves the entire scalar cone, so set $A_n = 1$. Then

$$0 \le B_n \le (M_n/M_c)^{\kappa-1} - 1 \longrightarrow 0, \qquad E_n \to 0, \qquad \mathcal{F}(U_n) \to \kappa.$$

Theorem 5.1 gives $U_n \to Qe_1$ strongly in H^s after translations and internal unitaries. A further unitary aligns the internal vector $\int Q(x)U_n(x)\,dx$ with the positive real first coordinate, without losing this convergence. Thus every imaginary component of Y_n is orthogonal to Q.

Equation (44) gives $\widetilde{\omega}_n \to \omega_*$, even if D_n is unbounded. Potential form continuity and (39) give

$$\langle Y_n, [a(D) + \widetilde{\omega}_n - V_{\rho_{U_n}}]Y_n\rangle \ge c\|Y_n\|^2_{H^s(\mathcal{H})}.$$

For large n, $M(Y_n)/M_n < 1/4$. All terms on the right of (43) are nonnegative. Since $E_n \le 0$, we obtain $E_n = 0$, $Y_n = 0$ and $D_n = 0$. The imaginary equation reduces to $v_n \cdot \nabla X_n = 0$. A nonzero L^2 function cannot be constant in a nonzero direction, so $v_n = 0$.

Every real component belongs to the kernel of $T_n = a(D) + b_n(D) + \omega_n - V_{\rho_{U_n}}$. Since $D_n = 0$, $\omega_n = \widetilde{\omega}_n$. This form is strictly positive on $Q^\perp$, by phase coercivity and $b_n \ge 0$. Its real kernel has dimension at most one, since two independent kernel vectors would have a nonzero combination in $Q^\perp$. Therefore all components are multiples of one real scalar stationary profile, the desired rigidity. The sequential contradiction proves a uniform δ. □

5.7 Polarization of low positive-energy traveling waves

For pure a, a common-frequency profile also satisfies the phase-boost identity

$$vM(U) = \int \nabla a(\xi)\|\widehat{U}(\xi)\|^2\,d\xi. \tag{45}$$

To justify it without moments, choose real $\chi \in C_c^\infty$ equal to x_j near zero, put $\chi_R = R\chi(x/R)$, and vary the profile action along $e^{i\theta\chi_R}U$. The density is unchanged. The commutator kernel contains $(a(\xi)-a(\eta))R^{N+1}\widehat{\chi}(R(\xi-\eta))$. The bound $|a(\xi)-a(\eta)| \le C|\xi-\eta|(|\xi|+|\eta|)^{m-1}$ yields a uniform form bound on H^s. For Schwartz fields the limit is Fourier differentiation; density and the uniform bound extend it to the energy space, giving (45) componentwise and after summation.

Proof of Corollary 5.3. If no uniform η existed, take unpolarized profiles with $M_n \to M_c$, $A_n = 1$ and $E_n \to 0$. Then $\mathcal{F}(U_n) \to \kappa$, so $U_n \to Qe_1$ strongly in H^s after symmetries. The right side of (45) converges to zero: Q is real, $|\widehat{Q}|^2$ is even, and ∇a is odd of order $m-1$. Thus $v_n \to 0$. Equation (42) gives $\omega_n \to \omega_*$. The complex self-adjoint frozen profile form

$$T_n = a(D) + \omega_n + iv_n \cdot \nabla - V_{\rho_{U_n}}$$

converges in form norm to L_Q; the drift has order one, strictly below m. Hence it is positive on the complex codimension-one space $Q^\perp$. Its complex kernel has dimension at most one. All internal components of U_n belong to this kernel, so they are constant complex multiples of a single scalar profile, a contradiction. □

Scope of the internal symmetry. A non-scalar matrix symbol or an internal rotation $u(t,x) = e^{it\Omega}U(x-vt)$ with a non-scalar Hermitian Ω is a different problem. In the latter case different components may occupy different spectral values of a common frozen spatial operator. The common-frequency kernel argument does not apply. The statements above also do not classify arbitrary time-dependent vector solutions.

6 Homogeneous transport, anisotropic dispersion, and nonlocal attraction

The transport argument extends to general homogeneous symbols without a sign assumption on the localized principal commutator. Its key is an exact cancellation on a spatially supported core. We obtain a common time-dependent result for local critical powers and Hartree attraction. The quantity localized in the general theorem is the full homogeneous kinetic density; the stronger gradient conclusion for isotropic quartic dispersion is considered separately at the end.

6.1 Equations and solution class

Fix $m > 1$, $N \geq 1$, and $s = m/2$. Let a be real, even, homogeneous of degree m, C^{N+2} away from zero, and positive elliptic:

$$c|\xi|^m \leq a(\xi) \leq C|\xi|^m.$$

Set

$$A(u) = \int a(\xi)|\widehat{u}(\xi)|^2\,d\xi, \quad M(u) = \|u\|_2^2, \quad P(u) = \int \xi|\widehat{u}(\xi)|^2\,d\xi.$$

Momentum is finite since $m > 1$. We treat either

$$iu_t = a(D)u - |u|^{p-2}u, \qquad p = 2 + 2m/N, \qquad E(u) = \tfrac{1}{2}A(u) - \tfrac{1}{p}\|u\|_p^p, \tag{46}$$

or, for $1 < m < N$,

$$iu_t = a(D)u - (K * |u|^2)u, \qquad E(u) = \tfrac{1}{2}A(u) - \tfrac{1}{4}H(u), \tag{47}$$
$$H(u) = \iint K(x-y)|u(x)|^2|u(y)|^2\,dx\,dy.$$

Here K is real, even, C^1 off zero, homogeneous of degree $-m$, and

$$|K(x)| + |x||\nabla K(x)| \leq C_K|x|^{-m}. \tag{48}$$

Positive type is unnecessary for the transport theorem: all nonlinear error estimates use the absolute Riesz interaction. This differs from the positivity needed for the phase-kernel argument.

Throughout this section, $u \in C([0,\infty); H^s)$ is a strong energy solution with conserved M, P, E, satisfying

$$\sup_{t\geq 0}\|u(t)\|_{H^s} \leq B < \infty. \tag{49}$$

We assume the solution is obtained, on compact time intervals, as a strong energy-space limit of sufficiently smooth solutions, so that the integrated localized virial identity follows from the estimates below. No local well-posedness theorem for arbitrary anisotropic symbols is asserted here.

Choose $\chi \in C_c^\infty$ with $0 \le \chi \le 1$, equal to one on $|x| \le 1$ and zero on $|x| \ge 2$, and put $\eta = 1 - \chi$. For a center $c(t)$ define

$$T_R^c(t) = \big\| |D|^s [\eta((x - c(t))/R) u(t)] \big\|_2^2, \tag{50}$$

$$D_R^c(t) = \int_{|x-c(t)|>R} \big| |D|^s u(t,x) \big|^2 \, dx. \tag{51}$$

These are distinct quantities, but the cutoff commutator below makes their localization properties equivalent on a bounded energy ball.

6.2 The kinetic commutator is exact on the core

Choose a real smooth compactly supported vector field f equal to x on $|x| \le 4$. Set

$$f_R(x) = Rf(x/R), \qquad L_R = f_R \cdot \nabla + \tfrac{1}{2} \operatorname{div} f_R, \qquad B_R = -iL_R, \qquad C_R = [a(D), L_R].$$

Thus L_R is skew-adjoint, while B_R and C_R are symmetric forms. Write

$$Q_R(w)^2 = \| |D|^s w \|_2^2 + R^{-m} \|w\|_2^2.$$

Lemma 6.1 (Uniform form bound and core cancellation)**.** *For all $R > 0$ and $z, w \in H^s$,*

$$|\langle z, C_R w \rangle| + |\langle z, ma(D) w \rangle| \le C Q_R(z) Q_R(w). \tag{52}$$

If z is supported in $|x| \le 2R$, then

$$\langle z, [C_R - ma(D)] z \rangle = 0. \tag{53}$$

Consequently, on a bounded H^s ball and for $R \ge 1$,

$$|\langle u, C_R u \rangle - mA(u)| \le C_B \left(\| |D|^s (\eta_R u) \|_2 + R^{-s} \right). \tag{54}$$

All statements also hold about any fixed center.

Proof. At $R = 1$ the Fourier kernel of C_1, up to a fixed normalization and the factor i, is

$$\tfrac{1}{2}[a(\xi) - a(\zeta)](\xi + \zeta) \cdot \widehat{f}(\xi - \zeta).$$

The homogeneous derivative bound gives

$$|a(\xi) - a(\zeta)| \, |\xi + \zeta| \le C |\xi - \zeta| (|\xi| + |\zeta|)^m.$$

After division by $\langle \xi \rangle^s \langle \zeta \rangle^s$, the right side is at most $C|\xi - \zeta| \langle \xi - \zeta \rangle^s$. Its product with $|\widehat{f}(\xi - \zeta)|$ is integrable. Schur's test therefore bounds $C_1 : H^s \to H^{-s}$. Under the unitary rescaling $v(y) = R^{N/2} w(Ry)$, C_R becomes $R^{-m} C_1$. This proves (52); the term $ma(D)$ is immediate.

For smooth compactly supported z in the stated ball, $L_R z = (x \cdot \nabla + N/2) z$. Homogeneity and differentiation of $A(\lambda^{N/2} z(\lambda \cdot)) = \lambda^m A(z)$ give

$$2 \operatorname{Re} \langle a(D) z, L_R z \rangle = mA(z),$$

which is (53). Approximation within a slightly larger ball and (52) extend the equality to supported H^s functions. This is equality of quadratic forms, not a claim that the defect operator annihilates z pointwise.

Let $z = \chi_R u$, $w = \eta_R u$ and $\mathcal{D}_R = C_R - ma(D)$. The core equality gives

$$\langle u, \mathcal{D}_R u \rangle = \langle w, \mathcal{D}_R w \rangle + 2 \operatorname{Re} \langle z, \mathcal{D}_R w \rangle.$$

The smooth-cutoff product estimate

$$\| |D|^s (\chi_R u) \|_2 \le C \big(\| |D|^s u \|_2 + R^{-s} \|u\|_2 \big)$$

and (52) now prove (54). The product estimate itself follows at scale one from the convolution inequality $|\xi|^s \le C_s(|\zeta|^s + |\xi - \zeta|^s)$ and Young's inequality, followed by rescaling. □

No favorable sign of the principal symbol $\nabla a(\xi)\cdot Df_R(x)\xi$ is used. In particular, the proof does not infer such a sign from ellipticity or convexity of a.

Lemma 6.2 (Cutoff energy and the literal kinetic tail)**.** *Put $\delta=\min(s,1)$. On the ball (49),*

$$(T_R^c)^{1/2}\le (D_R^c)^{1/2}+C_BR^{-\delta},\qquad (D_{2R}^c)^{1/2}\le (T_R^c)^{1/2}+C_BR^{-\delta}. \tag{55}$$

Proof. For $0<s<1$, the Fourier inequality $||\xi|^s-|\zeta|^s|\le|\xi-\zeta|^s$ and Young's inequality give

$$\|[|D|^s,\eta_R]u\|_2\le CR^{-s}\|u\|_2.$$

For $s\ge1$, use

$$||\xi|^s-|\zeta|^s|\le C_s|\xi-\zeta|(|\zeta|^{s-1}+|\xi-\zeta|^{s-1})$$

to obtain

$$\|[|D|^s,\eta_R]u\|_2\le C\left(R^{-1}\||D|^{s-1}u\|_2+R^{-s}\|u\|_2\right).$$

Only the compact part $-\chi_R$ contributes to the commutator. The uniform bound is $C_BR^{-\delta}$. Since $\eta_R=0$ inside the radius-R ball and $\eta_R=1$ outside the radius-$2R$ ball, the triangle inequality proves both assertions. □

6.3 The common time-dependent estimate

Let c be Lipschitz with $\|c'\|_\infty\le L$ and set

$$J_R(t)=2\langle u(t),B_R^{c(t)}u(t)\rangle=2\,\mathrm{Im}\int f_R(x-c(t))\cdot\overline{u}\nabla u. \tag{56}$$

For $s<1$, the last expression denotes the symmetric $H^{1/2}$ quadratic form, defined by density.

Lemma 6.3 (Two-sided localized virial estimate)**.** *Set*

$$\gamma=\min\left(\tfrac12,\tfrac1m\right),\qquad \beta=\min\left(\tfrac m2,1\right),\qquad \widetilde\beta=\min\left(\tfrac m2,\tfrac2m\right).$$

There are constants C_0,C_1 depending on the fixed equation, M,B,L and cutoffs such that, for $R\ge1$,

$$|J_R(t)|\le C_0R, \tag{57}$$

$$|J_R'-4mE+2c'\cdot P|\le C_1\left((T_R^c)^\gamma+R^{-\beta}\right), \tag{58}$$

$$|J_R'-4mE+2c'\cdot P|\le C_1\left((D_R^c)^\gamma+R^{-\widetilde\beta}\right). \tag{59}$$

The derivative assertions are understood as integrated identities with an error bounded by the integral of the displayed right side.

Proof. The scale-one first-order Fourier-Schur form estimate and rescaling give

$$|\langle u,B_Ru\rangle|\le C(R\||D|^{1/2}u\|_2^2+M).$$

Since $s>1/2$, this proves (57). The kinetic contribution to J_R' is $2\langle u,C_Ru\rangle$. Lemma 6.1 controls its error from $2mA(u)$.

For the local equation, direct integration by parts gives the nonlinear term

$$-\frac{2(p-2)}{p}\int \mathrm{div}\, f_R(x-c)|u|^p.$$

Its difference from $-4m\|u\|_p^p/p$ is bounded by $C\int_{|x-c|>4R}|u|^p$. Critical interpolation gives

$$\int_{|x-c|>4R}|u|^p \le \|\eta_R u\|_p^p \le CM^{m/N}T_R^c. \tag{60}$$

For the Hartree equation, write $\rho=|u|^2$. Symmetrization gives the nonlinear term

$$\iint \rho(x)\rho(y)[f_R(x-c)-f_R(y-c)]\cdot\nabla K(x-y)\,dx\,dy.$$

Its difference from $-mH(u)$ vanishes if both points lie in $B(c,4R)$. The Lipschitz bound on f_R and (48) dominate the error by the absolute Riesz interaction with at least one point outside that ball. For $q=4N/(2N-m)$, Hardy–Littlewood–Sobolev and interpolation yield

$$\begin{aligned}|\text{Hartree error}| &\le C\|\eta_R u\|_q^2\|u\|_q^2 \le C_{M,B}(T_R^c)^{1/2},\\ \|w\|_q^2 &\le C\|w\|_2\||D|^s w\|_2.\end{aligned} \tag{61}$$

Symmetrization supplies one factor $|x-y|$ at the diagonal, so HLS also justifies passage from a regularized kernel.

Motion of the center adds

$$-2c'(t)\cdot\int Df_R(x-c)\,\mathrm{Im}(\overline{u}\nabla u).$$

The difference from $-2c'\cdot P$ uses $g_R=Df_R-I$, which vanishes on $B(c,4R)$. Its current form on u is exactly its current form on $w=\eta_R u$: the cutoff is one on the support of g_R, and its derivative contributes only a real multiple of $|u|^2$. The rescaled first-order form estimate gives

$$\begin{aligned}\left|\mathrm{Im}\int g_R\overline{w}\nabla w\right| &\le C(\||D|^{1/2}w\|_2^2+R^{-1}\|w\|_2^2)\\ &\le C_M(T_R^c)^{1/m}+C_MR^{-1}.\end{aligned}$$

Here g_R is a constant matrix plus a rescaled smooth compactly supported matrix; its constant part is the ordinary momentum form. The last inequality is interpolation between orders zero and s. All current identities extend by density in $H^{1/2}$.

The principal terms are $4mE-2c'\cdot P$ in both equations. Since T_R^c is uniformly bounded, reducing the powers $1/2,1,1/m$ to γ proves (58). Lemma 6.2 gives

$$(T_R^c)^\gamma \le C(D_R^c)^\gamma + C_BR^{-2\delta\gamma}.$$

Since $2\delta\gamma=\widetilde{\beta}\le\beta$, this proves (59). Every form and nonlinear expression used above is continuous on bounded energy-space sets. Approximation therefore proves the integrated statements for the solution class specified above. □

6.4 Persistent exterior kinetic density and exact averaged drift

Theorem 6.4 (Transport at a distance proportional to the time window)**.** *Assume $E=-e<0$ and $P=0$. For every speed bound L, there are $\nu,d>0$, depending only on the fixed equation, M,B,e,L and cutoffs, such that every center c with speed at most L satisfies*

$$\liminf_{T\to\infty}\inf_{S\ge0}\frac{1}{T}\int_S^{S+T}\int_{|x-c(t)|>\nu T}\left||D|^{m/2}u(t,x)\right|^2dx\,dt\ge d. \tag{62}$$

Proof. Integrate the upper half of (59) over $[S,S+T]$ and use (57):

$$C_1\frac{1}{T}\int_S^{S+T}(D_R^c)^\gamma\,dt \ge 4me - C_1R^{-\widetilde{\beta}} - 2C_0R/T.$$

Choose $\nu>0$ with $2C_0\nu\le me$ and put $R=\nu T$, held fixed during this time integration. For large T, the right side is at least $2me$. Jensen's inequality, using $\gamma\le1$, gives the asserted lower bound with $d=(2me/C_1)^{1/\gamma}$. The bounds are uniform in S and in all centers of the given speed. □

Remark 6.5. The conclusion concerns the nonlocal homogeneous kinetic density when m is not an even integer. Its exterior integral is uniformly bounded, so the lower average forces a positive fraction of times in each long window to have nonzero exterior kinetic density. This is not a finite-speed theorem or a finite radiation-budget estimate, and does not exclude indefinite replenishment of exterior activity.

Theorem 6.6 (Exact drift under averaged kinetic localization)**.** *For either equation and a Lipschitz center c, suppose*

$$\lim_{R\to\infty} \limsup_{T\to\infty} \frac{1}{T}\int_0^T D_R^c(t)\,dt = 0. \tag{63}$$

Then the limit exists and

$$\lim_{T\to\infty} \frac{P\cdot[c(T)-c(0)]}{T} = 2mE. \tag{64}$$

In particular $E<0$ and $P=0$ exclude (63).

Proof. Integrate the two-sided estimate (59), divide by T, and use (57) and Jensen. For fixed R, the limiting discrepancy from (64) is at most

$$C\left[R^{-\widetilde{\beta}} + \left(\limsup_{T\to\infty}\frac{1}{T}\int_0^T D_R^c(t)\,dt\right)^{\gamma}\right].$$

Let $R\to\infty$. □

The drift theorem holds at every energy sign. For a traveling wave, it reproduces $v\cdot P = 2mE$. Its content for a general localized trajectory does not require a rigid profile or periodicity.

6.5 Uniform localization with an arbitrary center

Theorem 6.7 (No regularity assumption on the localization center)**.** *Suppose $E<0$ and an arbitrary map $b:[0,\infty)\to\mathbb{R}^N$ satisfies*

$$\sup_{t\ge 0} D_R^b(t) \longrightarrow 0 \qquad (R\to\infty). \tag{65}$$

Then b remains within a bounded distance of a Lipschitz center, and

$$\lim_{T\to\infty}\frac{P\cdot[b(T)-b(0)]}{T} = 2mE.$$

Consequently no solution obeying (49), $E<0$, $P=0$, and (65) exists.

Proof. Lemma 6.2 first gives uniform localization of T_R^b. In the local case take $z=p$; negative energy gives $\|u(t)\|_p^p \ge p|E|$. In the Hartree case take $z=q=4N/(2N-m)$. Since $H(u) = 2A(u)+4|E| \ge 4|E|$ and $|H(u)|\le C\|u\|_q^4$, the L^q norm also has a uniform positive lower bound. Estimates (60) and (61), applied at smaller cutoff radii, show that the L^z norm outside $B(b(t),2R)$ tends to zero uniformly in time.

The equation gives $\|u_t\|_{H^{-s}}\le C_B$. For the local term, this follows by Hölder and $H^s\hookrightarrow L^p$; for Hartree, use

$$\left|\int (K*|u|^2)u\overline{h}\right| \le C\|u\|_q^3\|h\|_q \le C_B\|h\|_{H^s}.$$

Choose σ strictly between $N(1/2-1/z)$ and s. Sobolev embedding and interpolation yield

$$\|u(t)-u(t')\|_{L^z} \le C_B|t-t'|^{(s-\sigma)/(2s)}. \tag{66}$$

Fix R_0 so that the exterior L^z norm is much smaller than the positive core norm, and then $\tau>0$ so that (66) is smaller still for $|t-t'|\le\tau$. If $|b(t)-b(t')|>4R_0$, the two balls of radius $2R_0$ are

disjoint. Restriction to the core ball at t' then contradicts (66). Hence $|b(t)-b(t')| \le 4R_0$ whenever $|t-t'| \le \tau$.

Linear interpolation through $b(n\tau)$ produces a center c with $\|c'\|_\infty \le 4R_0/\tau$ and $\sup_t |c(t) - b(t)| \le 8R_0$. A bounded displacement preserves literal kinetic-tail localization by inclusion of balls. Apply Theorem 6.6; the bounded displacement also does not affect the drift limit. □

The localization assumption allows a diffuse fixed-mass tail whose homogeneous kinetic norm tends to zero. Thus it is weaker than precompactness in H^s modulo translations. No uniform control of high-frequency tails is needed.

6.6 A stronger exterior-gradient theorem for quartic Hartree

For $a(\xi) = |\xi|^4$, $N > 4$, and a kernel satisfying (48) with $m = 4$, put

$$G_R^c(t) = \int_{|x-c(t)|>R} |\nabla u(t,x)|^2 \, dx.$$

Use the usual radial cutoff $f_R = \nabla \phi_R$ with $D^2\phi_R \le I$, equal to x on the core. The biharmonic kinetic term is bounded above by $8\|\Delta u\|_2^2 + O_B(R^{-2})$. In the Hartree estimate, $q = 2N/(N-2)$ and homogeneous Sobolev give

$$|\text{Hartree error}| \le C\|\eta_R u\|_q^2 \|u\|_q^2 \le C_B \|\nabla(\eta_R u)\|_2^2 \le C_B(G_R^c + R^{-2}).$$

The center-current error is at most $C\sqrt{M}L(G_R^c)^{1/2}$. Consequently

$$J_R' \le 16E - 2c' \cdot P + C\big((G_R^c)^{1/2} + R^{-2}\big). \tag{67}$$

Corollary 6.8 (Quartic nonlocal exterior-gradient persistence)**.** *For the quartic Hartree equation just specified, a forward-global uniformly H^2-bounded solution with $E < 0$ and $P = 0$ satisfies*

$$\liminf_{T\to\infty} \inf_{S\ge 0} \frac{1}{T} \int_S^{S+T} G_{\nu T}^c(t)\, dt \ge d > 0$$

for every bounded-speed center, with constants depending on its speed bound. Uniform gradient localization about an arbitrary center is impossible under those assumptions.

Proof. The time-window integration of (67) is identical to the proof of Theorem 6.4. For an arbitrary uniformly localizing center, the L^q lower bound, the Sobolev estimate on the exterior gradient, and the temporal continuity argument in Theorem 6.7 produce a bounded-speed center. Fixed-radius integration then contradicts $E < 0, P = 0$. □

This gradient conclusion uses the isotropic fourth-order kinetic sign; it is not claimed for arbitrary anisotropic a. The general two-sided theorem instead gives exact drift at the full kinetic-localization level.

6.7 Relation to earlier virial arguments

Localized virial and compact-orbit rigidity are established methods. Boulenger–Himmelsbach–Lenzmann, *Blowup for fractional NLS*, arXiv:1509.08845, develop the fractional localized virial framework. Dinh, arXiv:1808.07368, Proposition 1.5, already proves finite-time blowup or an unbounded kinetic subsequence for nonradial mass-critical isotropic dispersion of degree $1 < m < 2$ with negative energy, without a localization hypothesis. That specialization is therefore not a new open-problem resolution here. Cho–Hwang–Kwon–Lee, arXiv:1208.2302, give earlier negative-energy Hartree blowup results under their stated assumptions.

The extension established in this section is the common anisotropic all-degree transport argument, including nonlocal kernels, the exact averaged drift law and its kinetic-localization form. No priority claim is based solely on this comparison. Nor do these results exclude a bounded global orbit that indefinitely sustains the exterior activity required by Theorem 6.4.

7 Relation to prior work and the remaining dynamical problem

The variational stability argument follows Cazenave–Lions [2]. Nonhomogeneous sharp interpolation and constrained standing waves for mixed dispersion have been studied, for example, by Fernández–Jeanjean–Mandel–Mariş [3]. The specific object computed here is the limiting concentration energy for a bounded, possibly angular, high-frequency deficit, and the all-data critical-mass bound below it. A proof of this statement does not by itself establish its first priority.

Charge-response degeneracy, neutral Jordan chains, and Hamiltonian spectral bifurcation are part of established solitary-wave theory; see Grillakis–Shatah–Strauss [4], Comech–Pelinovsky [5], and Lin–Zeng [6]. The explicit small-velocity coefficient here includes the projection of the scaling derivative away from translations. Its real-profile nondegeneracy hypothesis remains a separate model-dependent question.

Bugiera–Lenzmann–Schikorra–Sok [7] treat boosted ground waves for general elliptic dispersion; their existence class includes the critical higher-order equation considered here. Their existence and symmetry conclusions do not supply the differentiable nondegenerate branch assumed in our spectral calculation. In the pure quartic critical case, Bonheure–Castéras–Gou–Jeanjean [8], Theorem 1.2, prove instability of radial standing ground waves by finite- or infinite-time blowup. Our nonzero-velocity spectral asymptotic addresses a different statement and does not replace that nonlinear result.

Sharp vector interpolation is also established; Ma–Zhao [9] derive such inequalities for coupled systems. Cheng–Guo–Hwang–Yoon [10] study scattering for finite and infinite coupled cubic systems. Our common scalar dispersion and total-density interaction give a full internal unitary symmetry. The polarization and compactness arguments depend on that symmetry; they make no claim about arbitrary matrix couplings or distinct internal frequencies.

Localized virial arguments are central in the work of Pausader and Shao [11], Boulenger, Himmelsbach and Lenzmann [12], and the mass-critical Hartree literature [14]. In particular, Dinh's Proposition 1.5 [13] already excludes a global uniformly bounded negative-energy orbit for the isotropic fractional mass-critical equation of order $1 < m < 2$, without a radial or localization assumption. The fractional specialization of our transport theorem is therefore not a new resolution of that question. The precise common transport identity derived here allows arbitrary elliptic homogeneous anisotropy, all $m > 1$, and the stated Hartree class.

The principal unresolved issue along the original homogeneous quartic line is still the near-threshold evolution of nonradial negative-energy data. Our profile gap and transport theorem do not exclude every persistent splitting or replenished radiation mechanism. Excluding bounded global evolution would also leave the separate possibility of infinite-time growth. Likewise, identifying a positive real spectral value does not supply a nonlinear instability proof for every symbol and dimension, and a static scale curvature does not by itself identify a Hamiltonian oscillation frequency. These are precise missing steps between the proved results and a complete classification of the flow.

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